Normalized remainder equations for ZFC consistency

Round 19 research report and reproducible artifacts

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Abstract

This report develops explicit Diophantine presentations whose unsolvability is provably equivalent in ZFC to syntactic ZFC consistency through identified inherited calibrations. A single normalized spherical remainder bound, a compatible odd-square radix, and certified monic eliminations give a balanced quartic in 244 unrestricted integers with 5212 collected monomials and coefficient height 3968. Companion equations trade dimension, sparsity and height. Polarizing same-logical square aliases removes equality dummies from coefficient-height-one quartics. A fixed-frame degree theorem identifies a precise obstruction to further scalar improvement by center changes alone. Complete new witness arguments, literal coefficient arrays, exact finite certificates and a reproducible offline package accompany the report. All domains and comparisons with Round 18 and the newest pinned inputs are explicit. No proof of consistency, global minimality or new end-to-end formal-kernel calibration is claimed.

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1 Results and interpretation

The main direct equation is now an explicit quartic in 244 unrestricted integer variables, with 176 quadratic residuals, 5212 collected monomials and coefficient height 3968. Compared with the newest incoming 252-variable common-remainder source, it improves all four varying measures. Compared with Round 18's 258-variable principal equation, it saves fourteen variables, ten residuals and 152 monomials while lowering coefficient height from 128446 to 3968.

The equation is fully specified by its finite residual array and its collected coefficient array. Appendix A prints all 176 residuals. It contains no variable exponent, factorial, binomial operator, unspecified coefficient or external predicate. Those notions occur only in the proof explaining its integer zeros.

Signed quartic	Variables	Rows	Terms	Height
Balanced	244	176	5212	3968
Sparse	250	182	4770	6709
Minimum selected dimension	243	175	19517	7680
Lower height dimension alternative	246	178	21537	6144
Lowest selected height	247	179	10422	3840

All five have exact degree four on $\mathbb{Z}$. The 243-variable choice is much denser, and the height-3840 choice nearly doubles the balanced support. They are genuine tradeoffs, not simultaneous improvements in every metric. The sparse choice retains 4770 terms while improving variables, rows and height over the incoming Shared bounds principal.

The new height-one transformation changes prescribed square labels to products of two aliases for the same logical value. It then replaces a three-coordinate equality cycle by a two-coordinate quartic equality penalty. A complete nonnegative energy argument preserves the source zero set; full coefficient collection, including all equality terms, certifies which particular polarizations retain height one.

The selected height-one variable/support pairs are

$$(528, 20816), \quad (534, 18543), \quad (580, 12001), \quad (620, 8318), \quad (717, 7176).$$

The 534-variable point improves both dimension and support over Round 18's 571-variable compact point; the 528-variable alternative trades larger support for six fewer coordinates. Matched-source replays separately isolate the effect of polarization from the new arithmetic source.

The scalar analysis distinguishes a limitation of the inherited frame from an unrestricted lower bound. With every other register fixed, no nonzero constant radical weight and integral polynomial center can reduce its output degree below 300. This theorem concerns that exact family only, not all eleven-variable polynomials or other arithmetic frames.

The logical meaning is a correspondence with a finite syntactic ZFC refutation, not an independent assertion that the equations have no roots. Each new witness implication is proved in both directions. The ordinary calibration is retained explicitly; finite coefficient checks and successful program execution do not become a new proof-assistant kernel theorem.

2 Logical scope and literal polynomial definitions

2.1 What the consistency interpretation asserts

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}$ denotes the unrestricted integers. The domain is part of an equation, not a presentation detail. An equation over $\mathbb{N}$ is not thereby an equation over $\mathbb{Z}$, and an unrestricted construction requires its own sign argument or an explicit domain conversion.

Let $\mathrm{Ref}_{\mathrm{ZFC}}(r)$ be the fixed arithmetized predicate saying that r codes a finite syntactic ZFC refutation. For each selected endpoint P on the stated domain D, the claim is

$$\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} \; \mathrm{Ref}_{\mathrm{ZFC}}(r). \tag{1}$$

The ordinary transformations proved here are composed with the identified inherited calibration. On that proof chain, formalization in ZFC yields

$$\mathrm{ZFC} \vdash [\forall \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) \neq 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{2}$$

This report does not assert that either sentence in (2) is true. In particular it does not prove consistency, decide solvability, establish a global minimum, or supply a new end-to-end proof-assistant kernel check or encoded object-language ZFC derivation.

For the direct families, the retained ordinary logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The first relates the fixed numerical calculus to a finite ZF refutation; the second is the retained ZF-to-ZFC refutation equivalence. The arithmetic front implements that calculus using finite syntax constructors, reciprocal CRT carriers, factorial and power certificates, and range tests. The scalar family instead retains the separately identified source34 native/compiler and refutation calibration, followed by its signed-rank and interpolation reductions. These are two calibrated routes, not interchangeable coefficient tuples.

2.2 The modular structure of the proof

The proof chain has four logically different layers.

1. The inherited calibration identifies a fixed finite arithmetic relation with syntactic refutation. Its provenance and exact numerical payloads are part of the source closure.

2. Unbounded arithmetic arguments prove both witness directions for the certificate relations. These include Pell classification, CRT choices, factorial fixed points, integer signs, and rank or interpolation lemmas.

3. Exact polynomial transformations replace one finite presentation by another. Integral substitutions, restored monic coordinates, graph identities and coefficient collection establish the claimed zero-set correspondence.

4. Finite independent checkers verify the actual emitted arrays and graphs, not merely an abstract instance of a theorem.

A hash certifies file identity, not a theorem; an exhaustive coefficient check certifies an identity, not an unbounded Pell assertion. Each layer is used only for what it establishes. The new sections give complete arguments for this round's transformations; the retained source closure supplies the inherited arithmetic and logical inputs at their stated boundary.

2.3 Explicit finite arrays define the direct equations

For a direct endpoint, the ordered `variables` array in `equation.json` fixes the independent coordinates $x_0, \ldots, x_{n-1}$. A term encoded as $[c, [i_1, \ldots, i_d]]$ denotes $c \prod_{j=1}^{d} x_{i_j}$; the empty index list denotes the constant c. Repeated indices are retained and give ordinary powers. The m residual rows $R_0, \ldots, R_{m-1}$ have integer coefficients and degree at most two. The equation is exactly

$$P(\boldsymbol{x}) = \sum_{j=0}^{m-1} R_j(\boldsymbol{x})^2 = 0. \tag{3}$$

Every coefficient and every coordinate appears in a finite supplied array. The adjacent `expanded_polynomial.json` stores the fully collected polynomial, including its constant term. There is no factorial, variable exponent, division, binomial coefficient, bounded quantifier, or sequence symbol among its algebraic primitives. Such expressions occur only in the proof explaining its zeros.

For either domain $D \subseteq \mathbb{Z}$, equation (3) is equivalent to $R_j(\boldsymbol{x}) = 0$ for every j, because each square is nonnegative. Exact degree four is established by a collected nonzero quartic coefficient, not merely by the upper bound $\deg R_j \leq 2$. All independent coordinates are counted; named arithmetic abbreviations are not new unknowns.

The height-one endpoints are defined directly by their collected coefficient arrays. Their graph certificates prove the displayed energy identity and its zero-set correspondence. The independent checker reconstructs the energy and compares every collected coefficient. The height claim therefore concerns the final polynomial, not only its generating rows.

2.4 Arithmetic graphs define the scalar equations

The scalar graph leaves are eleven independent unrestricted integer variables, ordinary integer constants, or six fixed authentic numerical ports. Each internal node is binary addition, subtraction, or multiplication. An acyclic graph with one designated output therefore denotes a unique element of an ordinary polynomial ring over $\mathbb{Z}$. Registers abbreviate polynomials; they do not introduce hidden existential variables. There is no division or radical gate.

The source34 tuple is the inherited `mixed-parity-unrounded-1064921` tuple at coefficient source commit

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

Its ports include $\beta = 2^{1064921} - 1$, cone_A $= 2^{1113}$, cone_B $= 2^{666104}$, and the complete literal K_0, M_C, M_F payloads. The retained M_F already includes its native-value-plus-β shift. It must not be shifted twice or replaced by a port from a positive-domain tuple. The payload files contain complete minimal unsigned little-endian integers and are bound by byte length, bit length and SHA-256; neither a hash nor an ellipsis is substituted for a coefficient. The finite inherited compiler recipe reconstructs those values.

2.5 What each measure counts

For a nonzero collected polynomial $P = \sum_\alpha c_\alpha x^\alpha$, write

$$\deg P = \max_{c_\alpha \neq 0} |\alpha|, \qquad |\operatorname{supp} P| = \#\{\alpha : c_\alpha \neq 0\}, \qquad \operatorname{ht}(P) = \max_\alpha |c_\alpha|.$$

Support includes a nonzero constant coefficient. The number of residual rows is distinct from support and from the number of operations. Scalar cost counts live binary graph gates, including squares and multiplication by fixed constants; copying a register and reading a leaf are free. Dead nodes do not count. Unless an expanded scalar polynomial is materialized, its support and total coefficient height are unmeasured, not zero.

A point dominates another only when it is no worse in every displayed measure and strictly better in at least one, with domain and degree held fixed where required. This is a finite comparison among specified presentations. Height one, smaller dimension, sparse support and low graph cost are different objectives, and no single global simplicity ordering is claimed.

3 Small signed quartics and their arithmetic boundary

Every variable in this section ranges over $\mathbb{Z}$. The new finite coordinate and row transformations are applied to an authenticated signed arithmetic source, not to a natural-number equation with its signs silently discarded. This section first supplies the unbounded argument needed by that source; then it proves the exact algebraic transformations used in the new endpoints. The source files and each frozen transformation are part of the release.

3.1 The unchanged signed arithmetic interface

Write t, F, B, V, y, x, N, q for the semantic coordinates before integral centering. The unchanged signed Pell frame and semantic anchor identify both coordinates of the principal Pell pairs before any low-digit sign is used. This is the orientation-first interface retained from Round 17. Its signed power lemma says that a framed power request at an integer base $n \geq 2$ forces its requested exponent k to be nonnegative and its output to equal n^k. Briefly, the classified signed Pell index gives the power congruence; if $k < 0$, multiplication by n^{-k} and the strict frame bound would force $mn^{-k} = 1$ for an integer output m, which is impossible. For base one and a known nonnegative index, the same congruence and frame bound give output one. This argument does not assume the output is positive.

The retained ceiling signs and endpoint rows give

$$\begin{aligned} t &\geq 16, & V &= (F^2+1)^t, \\ FB &= V + s_0, & B &= r_0 + s_0 + 1, \qquad r_0, s_0 \geq 0, \\ y &= V(V-1), & x &= F + 1 + (y+1)u_0, \\ x &= 1 + [1 + (t+1)y]u_t, & N &= B^x, \\ H_0 &= 1 + \sum_{i=1}^{9} a_i, & q &= (BH_0)^x. \end{aligned} \tag{4}$$

The powers here describe the proved meanings of existing quadratic certificates; exponentiation is not an operation in the final polynomial. The a_i are the actual encoded representatives. The derived constructor label h satisfies the retained ledger bound

$$0 < h \leq (n_* + 1)^\ell \leq (BH_0)^\ell, \qquad n_* = \max(1, a, b, c, d), \quad \ell = 2^{22}, \tag{5}$$

once $a_i \geq 0$ and $B \geq 2$. The last inequality follows from $n_* + 1 \leq 2H_0 \leq BH_0$, including the case $H_0 = 1$.

3.2 An acyclic odd square bootstrap

The odd-square source replaces the old ternary guard on $q - B - 1$ by

$$q = (2z+1)^2. \tag{6}$$

Only $q \geq 1$ follows initially. The unchanged orientation argument uses the square frame, not this missing sign. The request for V has known index $t \geq 16$ and base at least one. Thus $V = (F^2+1)^t \geq 1$. The ceiling equations force $F \geq 1$ and $B > F$, since $FB \geq V \geq F^2 + 1$ while $B \geq 1$. In particular $B \geq 2$ and $y > F + 2$.

The genuine request $N = B^x$ now forces $x \geq 0$. If $u_0 \leq -1$, the first endpoint would give $x \leq F - y < 0$; hence $u_0 \geq 0$ and $x \geq F + 1 \geq 2$. The last endpoint then forces $u_t \geq 1$, so

$$x \geq 2 + (t+1)y. \tag{7}$$

The numerator requests at base $q + 1 \geq 2$ force all nine $a_i \geq 0$; the derived-label request and nonzero anchor give $h > 0$. Consequently $H_0 \geq 1$, and the existing request for q yields $q = (BH_0)^x \geq B^2 > B$. The deleted output slack is therefore $q - B - 1 \geq 0$. Every original front sign has now been recovered, without using a low-digit remainder sign or the conclusion $F = t!$.

3.3 A uniform coefficient bound before factorial decoding

The useful margin can be proved entirely before the factorial fixed point. Since $V \geq 2^t$, for $t \geq 16$ we have

$$y \geq 2^t(2^t - 1) > \ell, \qquad y \geq t + 2.$$

Combining these integer inequalities with (7) gives

$$x \geq \ell t + 4, \qquad x \geq t(t+1) + 4. \tag{8}$$

All requested digit positions have $k \leq x$. For every coefficient below the requested position, the following three estimates apply.

For an ordinary upper index $0 \leq a < H_0$ and $0 \leq j < k \leq x$,

$$\binom{a}{j} \leq \max(1, a)^j \leq H_0^x \leq q/16,$$

because $B \geq 2$ and $x \geq 4$. For the derived label, $k = t + 1$, so

$$\binom{h}{j} \leq h^j \leq (BH_0)^{\ell t} \leq (BH_0)^{x-4} \leq q/16 \qquad (0 \leq j \leq t).$$

For the factorial upper index, $F^2 + t \leq B^{t+1}$: indeed $F^2 < B^2 \leq B^t$, $t \leq 2^t \leq B^t$, and $2B^t \leq B^{t+1}$. Therefore

$$\binom{F^2 + t}{j} \leq (F^2 + t)^t \leq B^{t(t+1)} \leq B^{x-4} \leq q/16 \qquad (0 \leq j \leq t).$$

These inequalities assume no factorial identity. Put

$$\rho(a, k) = \sum_{j=0}^{k-1} \binom{a}{j} q^j, \qquad D = q^k.$$

For every needed pair (a, k), the estimates give

$$0 \leq \rho(a, k) \leq \frac{q(D-1)}{16(q-1)} \leq \frac{D-1}{8} < \frac{D}{8}. \tag{9}$$

3.4 Recovering all remainder signs from shared budgets

There are eleven remainders. The two with denominator q^t are the factorial digit remainder and encoded-index remainder; five have denominator q^{t+1}, including the derived-label remainder; two stack remainders have denominator q^{y+1}; and two strict quotient remainders have denominator q^x. The old interval guards have the form

$$D = r + s + 1, \qquad rs = u^2 + v^2 + w(w+1).$$

The second expression is nonnegative at every integer tuple and represents every nonnegative integer. To prove the latter assertion, apply the three-square theorem to $4n + 1$. Modulo four exactly one of its three square summands is odd. Write it as $(2w+1)^2$ and the even squares as $(2u)^2, (2v)^2$, then divide by four.

For the symmetric version, retain the linear interval rows and replace a group's private product guards by

$$D^2 - 4\sum_{i \in G} r_i^2 - 1 = u^2 + v^2 + w(w+1). \tag{10}$$

Every new zero has $|r_i| < D/2$. The unchanged numerator row gives $r_i \equiv (q+1)^{a_i} \equiv \rho(a_i, k) \pmod{D}$. For the factorial row its other part is $D(B + qH)$; for a range row it is DNQ. Both are multiples of D regardless of the signs of H, Q. Thus

$$|r_i - \rho(a_i, k)| < D/2 + D/8 < D,$$

so the divisible difference is zero. All remainders are canonical and nonnegative, and $s_i = D - 1 - r_i \geq 0$ restores every old interval guard. For the factorial digit, subtract its recovered tail and divide by q^t. The quotient identity gives

$$B \equiv \binom{F^2 + t}{t} \pmod{q}.$$

The displayed coefficient bound also holds at $j = t$, so the binomial is in $[0, q)$; the recovered front guard gives $0 < B < q$. Thus $B = \binom{F^2+t}{t}$ exactly. Only after this recovery is the inherited factorial fixed-point theorem used to deduce $F = t!$. The unchanged CRT range and local Horn arguments then produce an actual finite C222 derivation and its inherited ZFC refutation interpretation.

Conversely, construct the inherited canonical witnesses from a finite refutation. A group of at most five remainders obeys $4\sum \rho_i^2 < 5D^2/16 < D^2$. The nonnegative integer $D^2 - 4\sum \rho_i^2 - 1$ therefore has a ternary representation, supplying all new auxiliaries.

The adjacent levels merge by adding two product coordinates $z_0 = qr_0$, $z_1 = qr_1$ and using one group at $D' = q^{t+1} = qD$. Its seven entries are z_0, z_1 and the five higher-level remainders. Soundness gives $|r_i| < D/2$ by division by $q > 0$; completeness uses $4\sum \rho_i^2 < 7D'^2/16 < D'^2$. The two product rows are quadratic. Replacing two ternary triples by one while adding two products saves one coordinate. The alternative spherical version uses $D^2 - 2\sum r_i^2 = u^2 + v^2 + w(w+1)$; it gives $|r_i| \leq D/\sqrt{2} < 3D/4$, still enough because $3D/4 + D/8 < D$. Its seven-entry completeness bound is $2\sum \rho_i^2 < 7D^2/32 < D^2$. These are two different literal budget formulas, each with a complete signed uniqueness argument.

3.5 Completeness of the odd square restriction

To construct (6), choose the beta code x even. In true witnesses, $F = t!$, $V = (F^2+1)^t$, and $y = V(V-1)$ is even. Every beta CRT modulus $1 + (i+1)y$ is odd; adding their odd common modulus changes the code's parity. A sufficiently large even representative preserves all endpoint and nonnegative quotient requirements.

Each exact factorial-product factor $1 + F^2/i$ is odd, since $F^2/i = F(F/i)$ is even for $1 \leq i \leq t$. Hence B, and then $N = B^x$, are odd. Each of the nine independent CRT representatives can be shifted by the odd modulus N^2, so their parities can be prescribed independently. Choose opposite parities for a, b as required by the semantic anchor, and choose one of the other seven parities so that their total sum is even. Then H_0 is odd. All ordinary representatives can retain their old bounds at most $2N^2$; choose the dominant a in $[2N^2, 4N^2)$ with its prescribed parity. The inherited dominance and frame estimates are therefore preserved. Finally BH_0 is odd and x is even, so

$$z = \frac{(BH_0)^{x/2} - 1}{2} \in \mathbb{Z}$$

supplies the odd-square row. This is a construction of fresh witnesses, not a claim that every old zero already has an odd-square radix.

3.6 One normalized budget across all four denominator levels

The incoming Common remainders construction uses a stronger sharing step. Set $M = q^x$ and let L_i be each remainder's original denominator. By (7), every exponent defining L_i is at most x; hence L_i is a positive divisor of M. For the three smaller denominator levels introduce one ratio coordinate each, with a quadratic row

$$Lv_L = M.$$

For the largest level take $v_M = 1$ without adding a coordinate. Replace an old digit identity $U_i = L_iQ_i + r_i$ by

$$v_{L_i}U_i = MQ_i + R_i. \tag{11}$$

All displayed high quotients are existing registers or products arranged as in the literal quadratic graph. In particular, a range digit's old product NL_i can be replaced by the shared product NM, so the new actual row is $v_{L_i}U_i = (NM)H_i + R_i$. The factorial digit uses the existing register for $B + qH$. No cubic product is silently inserted into a residual.

Lemma 3.1. *The scaled identities* (11)*, the ratio rows, and*

$$M^2 - 2\sum_{i=1}^{11} R_i^2 = u^2 + v^2 + w(w+1) \tag{12}$$

recover every old remainder interval. Conversely, canonical witnesses with $0 \le r_i < L_i/8$ *extend to these rows.*

Proof. Each ratio equals $M/L_i > 0$. Since both $v_{L_i}U_i$ and MQ_i are divisible by v_{L_i}, so is R_i. Thus $r_i = R_i/v_{L_i}$ is an integer; this is a proof-only division, not a gate or an extra rational witness. Dividing the scaled row restores the original digit identity. Equation (12) gives $|R_i| \le M/\sqrt{2} < 3M/4$, hence $|r_i| < 3L_i/4$. By (9) and congruence, the difference $r_i - \rho(a_i, k_i)$ is divisible by L_i and has absolute value below $7L_i/8 < L_i$. Hence it is zero. All deleted interval slacks and product witnesses can be restored exactly as before.

Conversely, define $v_{L_i} = M/L_i$ and $R_i = v_{L_i}r_i$ from canonical witnesses. Every scaled row holds, and $0 \le R_i < M/8$. Therefore

$$2\sum_{i=1}^{11} R_i^2 < \frac{11}{32}M^2 < M^2.$$

The integer $M^2 - 2\sum R_i^2$ is nonnegative and has the required ternary representation. This proves existence in both directions. □

The incoming common-denominator source instead uses $M^2 - 4\sum R_i^2 - 1 = G$. Its soundness radius is $M/2$, and the same canonical witnesses complete it because $4\sum R_i^2 < 11M^2/16 < M^2$. The two budgets are therefore existentially equivalent through the calibrated source relation, not by an asserted pointwise equality of their private ternary witnesses.

The new form combines the common-denominator idea with the earlier pre-factorial margin and, where specified, the compatible odd-square radix. In particular, the ratio rows are interpreted only after positive powers have been established; the large common guard is never used to establish the signs on which its own interpretation depends.

3.7 Certified integral changes preserve the entire zero set

Lemma 3.2. *Let* $R = (R_1, \dots, R_m)$ *be integer polynomials. Each of the following operations preserves the indicated integer zero set: an integral elementary coordinate shear; an elementary integral residual-row operation; and elimination of a coordinate by a monic defining row. The first two are bijections on the complete zero sets; the third is a bijection after restoring the eliminated coordinate by its defining polynomial.*

Proof. For a shear $x_i = y_i + cy_j$ with $i \ne j$ and $c \in \mathbb{Z}$, leave every other coordinate unchanged. The inverse is $y_i = x_i - cx_j$, again integral. Substitution gives $R(x) = R(Ly)$ identically. Translation by an integer constant has the equally explicit inverse translation.

For a row operation replace R_i by $R_i + cR_j$, with $i \ne j$, keeping all other rows. A zero of the old rows is a zero of the new rows. A zero of the new rows has $R_j = 0$, hence $R_i = (R_i + cR_j) - cR_j = 0$. Equivalently, the residual vector is multiplied by an elementary matrix in $\mathrm{GL}_m(\mathbb{Z})$. Its sum of squares generally changes, but its zero set does not.

For a row $x_i - f(x_{\ne i}) = 0$ with $f \in \mathbb{Z}[x_{\ne i}]$, substitute $x_i = f$ in every remaining row. A new zero extends uniquely by this value, and every old zero has that value. This proves both directions and uniqueness. An exact checker verifies that any degree restriction survives the substitution; it is not a general property of monic elimination. □

A frozen sequence of these operations is a finite mathematical certificate. The particular heuristic that found it is immaterial to its validity. A shear transported between two sources by coordinate names requires checking the full named substitution on the actual target rows; positional index agreement alone is insufficient. Residual height and support are measured only after collecting the whole final sum of squares.

3.8 The exact five selected endpoints

The starting literal array is the incoming centered Common remainders 252-variable, 182-row equation. Its translation certificate identifies the semantic radix coordinate. The new construction deletes exactly the old front guard on $q-B-1$, adds the odd-square row, and changes the common budget from coefficient four with constant -1 to coefficient two with constant zero. The two witness arguments above justify these semantic changes. The discarded private guard coordinates are absent afterward; all other rows are retained exactly at this stage. The resulting source has 250 variables and 182 rows.

The balanced construction first applies 76 fixed integral coordinate shears, transported by coordinate identity from the Shared bounds certificate. Five elementary residual additions expose three affine defining rows. In the intermediate 250-coordinate order, with zero-based indices and coordinates ξ_i, their unique restorations are

$$\begin{aligned}\xi_{37} &= -4 - 7\xi_{10} - 4\xi_{11} - 4\xi_{21} + \xi_{22},\\ \xi_{89} &= -12 + 6\xi_{3} + 8\xi_{84} + 4\xi_{85} + \xi_{86},\\ \xi_{166} &= -3 + 2\xi_{107} - 2\xi_{148} + \xi_{164}.\end{aligned}$$

The defining rows have unit coefficients on the eliminated coordinates. Substituting these formulas and deleting those rows reduces the system to 247 variables and 179 rows. After reindexing, 33 further elementary residual additions give the intermediate `balanced247` checkpoint. Every identity is certified against the actual preceding stage, and every remaining residual is still quadratic.

Three further monic quadratic rows can be eliminated without raising residual degree. In the 247-coordinate checkpoint order, write the coordinates as η_i. Their exact restorations are

$$\begin{aligned}\eta_{62} &= 10 + 14\eta_{53} + 4\eta_{53}^2,\\ \eta_{64} &= -\eta_{12} + \eta_{12}^2 + 4\eta_{12}\eta_{63} + 4\eta_{63}^2,\\ \eta_{45} &= 2\eta_{10} - \eta_{10}^2 - 4\eta_{10}\eta_{43} + 2\eta_{43} - 4\eta_{43}^2 + \eta_{44}.\end{aligned}$$

These are the semantic coordinates named `mul.281`, `add.292` and `mul.194`, respectively. At each deletion the coordinate occurs only as a linear monomial in every other residual. Such a residual has the form $av+G$ with $a \in \mathbb{Z}$ and G quadratic and independent of v; replacing v by its quadratic restoration keeps degree at most two. The independent checker verifies this occurrence condition before every quadratic deletion, in addition to checking all resulting degrees. Unique restoration proves the same zero-set bijection as Lemma 3.2. After these three deletions and nine further row additions, the result has 244 variables, 176 rows, 5212 terms and height 3968.

The 243-variable alternative also eliminates the coordinate `add.380` by a 15-term affine restoration, before those three quadratic deletions. In the same checkpoint order that additional formula is

$$\begin{aligned}\eta_{103} = 5\eta_{10} - 3\eta_{11} + 4\eta_{21} + 2\eta_{24} + 4\eta_{26} + 2\eta_{29} + 2\eta_{31} + 2\eta_{34}\\ + 2\eta_{36} + 4\eta_{38} + 2\eta_{39} + 2\eta_{42} + 2\eta_{46} + 4\eta_{67} + 2\eta_{68}.\end{aligned}$$

Its final frozen residual basis uses 802 further row additions. The sparse form instead uses the original 76 shears and seven residual additions, without any projection. The separate 246-variable form uses six elementary linear-coordinate shears, 1057 row additions and four affine projections.

The low-height form retains the first three affine projections and a longer frozen row basis. No optimization search is needed to define or reproduce any of these finite sequences.

Endpoint	Variables	Rows	Terms	Height
`balanced244`	244	176	5212	3968
`dimension243`	243	175	19517	7680
`sparse250`	250	182	4770	6709
`dimension246`	246	178	21537	6144
`lowheight247`	247	179	10422	3840

Every equation is the sum of the squares of its full residual array over $\mathbb{Z}$. The first four contain the collected monomial $13x_0^4$; the last contains $22x_0^4$. Since every row has degree at most two, these nonzero coefficients establish exact degree four. Full coefficient collection gives the displayed support and height, rather than an upper bound from the largest residual coefficient. The intermediate `exposed247` and `balanced247` arrays remain technical source and construction checkpoints; they are not presented as additional selected frontier points.

The inverse algebraic witness maps are explicit: reverse the residual operations, restore each eliminated coordinate, and reverse the coordinate shears. For the two semantic source changes, restore old guards with fresh ternary witnesses after the proven residue and sign recovery. Conversely, start with a finite refutation, use the compatible even-code and odd-radix canonical witnesses, fill the common spherical budget, then apply the exact coordinate maps. This composes both existential directions all the way to the retained calibrated source. The semantic stages are not claimed to give a bijection between arbitrary private witness choices.

4 Height one quartics with polarized square aliases

The source compiler acts on a finite quadratic system over unrestricted integers. Round 18 supplied certified normalization, row-sign balancing and opposite-occurrence fusion: equal monomials with opposite signs become a single incidence edge between their source rows. The new transformation acts later, on square aliases and their equality penalties. We first give the complete energy and witness argument, then prove the additional polarization step. Every height claim concerns the entire collected polynomial, including all equality penalties.

4.1 Certified normalization and row orientations

Let the given source be $R_r(x) = 0$. We first use the preceding exact affine refactorization, when applicable: if the homogeneous quadratic part of a row is $\eta L(x)^2$, with $\eta = \pm 1$, introduce $z = L(x) + k$ and replace the row by $\eta z^2 + R_r(x) - \eta(L(x) + k)^2 = 0$. The new definition is monic and its value is uniquely determined. Shared affine definitions can be reused. Coefficient-by-coefficient substitution verifies every transformed row. The provided original source, transformed source and factor certificate fix this operation completely.

Normalize integer coefficients by monic definitions of integer constants $t_c = c$ and scale variables $s_{c,i} = cX_i$. A constant is defined from already defined smaller constants as $t_c = t_a + t_b$ when $a + b = c$, using $t_1 = 1$. If no such pair is available, introduce the smaller binary intermediate and use $t_c = t_{\lfloor c/2 \rfloor} + t_{\lfloor c/2 \rfloor} + (c \bmod 2)$. Each added definition is triangular and monic. The delivered DAG prefers two already-needed constants; in particular it does not introduce a binary intermediate merely because a binary-only recipe would do so.

For each absolute quadratic coefficient $c > 1$, form a finite graph on the source coordinates, with an edge $\{i, j\}$ for every term cX_iX_j. A loop at i requires selecting i. Choose any vertex cover V_c of this graph, and introduce $s_{c,i} = t_cX_i$ only for $i \in V_c$. A term cX_iX_j is then represented by $s_{c,i}X_j$ if i was selected, or by $X_is_{c,j}$ otherwise. The cover condition proves that one option always exists. The retained deterministic greedy cover starts with the loop vertices and repeatedly selects

a highest-degree remaining vertex; optimality is not assumed. Constants and linear terms use t_c and $t_c X_i$, respectively. Unit coefficients need no constructor.

Splitting repeated summands into separate occurrences yields

$$S_r(y) = \sum_j \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every normalized coordinate has a unique integral value over an original tuple. The checker verifies all triangular pivots and each exact source-row pullback, rather than trusting the constant-chain or cover-selection recipe. Thus normalization preserves both witness maps. The coefficient-factor choice, just like the later graph, is fixed by the certificate and is not a parameter of the polynomial.

Choose a fixed sign $\sigma_r \in \{-1, 1\}$ for each normalized row, and replace S_r by $T_r = \sigma_r S_r$. This preserves both its zero set and uniqueness of all constructors. The choice is part of the finite certificate, not a parameter of the final polynomial. The supplied heuristic flips a row when doing so reduces

$$\sum_m \left| \sum_{r,j:m_{rj}=m} \sigma_r \epsilon_{rj} \right|.$$

This balances opposite occurrences of the same logical monomial. Correctness does not require optimality of this heuristic.

4.2 Direct edges for cancelling occurrences

Create disjoint vertex blocks B_r for the rows and a hub block B_0. Inside each block choose a tree, with one independent unrestricted integer flow variable on each tree edge. A normalized occurrence is assigned to a vertex in its own row block. Partition the occurrences into sets of size one or two. A pair is allowed only when its occurrences are in distinct rows, have the same logical monomial, and have opposite signs in T_r.

A singleton of sign ϵ is represented by an edge from its row vertex to a hub vertex carrying ϵM. A pair $+m$ in row r and $-m$ in row s is represented by one edge from its row-r vertex to its row-s vertex, carrying M. Here M is a degree-two homogeneous monomial in aliases of the logical factors of m: a linear term is multiplied by a unit alias and a constant uses two unit aliases. Each factor alias has one declared logical value. The two occurrences of a pair use *the same* edge and the same factor aliases.

The graph is required to be simple. The constructor rejects a pairing if its proposed unordered vertex pair was already used; remaining hub edges are placed without parallel edges. The certificate lists every bound source occurrence, and the checker verifies that each appears exactly once. More precisely, the edge boundary at the row-block level equals its list of signed source occurrences. Consequently, after substituting the intended logical values,

$$\sum_{v \in B_r} b_v = T_r(y), \qquad \sum_{v \in B_0} b_v = -\sum_r T_r(y). \tag{13}$$

where b_v is the divergence from prescribed edges alone. A fused edge contributes $+M$ and $-M$ to precisely its two bound rows; hence (13) is unchanged by fusion. This is the inherited occurrence-boundary identity.

4.3 The coefficient-height-one energy

Let d_v denote the outgoing-minus-incoming sum of all prescribed and free edges, and put

$$E = \frac{1}{2} \sum_v d_v^2.$$

Each edge has two endpoints, so its diagonal energy coefficient is one. Two distinct edges of a simple graph meet at most once, so their formal cross coefficient is 0 or ± 1. Give each prescribed

product an initial private factor alias, and share its other logical factor as in the previous incidence compiler. The initial fully expanded energy has height one. Subsequently identify only aliases of the same logical value, and accept an identification only when full exact coefficient collection still has coefficients in $\{-1, 0, 1\}$. Such substitution preserves the sum-of-squares identity even when signed collisions cancel.

For every remaining non-unit logical class with at least three aliases use

$$C(a_1, \dots, a_k) = \sum_i a_i^2 - \sum_i a_i a_{i+1}, \quad a_{k+1} = a_1.$$

Since $2C = \sum_i (a_i - a_{i+1})^2$, it is nonnegative and vanishes exactly when all aliases agree. A singleton needs no penalty. A class with two aliases is either completed to a three-cycle by one dummy, or uses

$$Q(a, b) = a^4 + b^4 - a^3 b - a b^3 = (a - b)^2 (a^2 + ab + b^2).$$

This is nonnegative and vanishes exactly at $a = b$; also $2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2$. The latter option is used only when neither alias occurs as a prescribed square, which ensures its four monomials cannot collide with the incidence energy. For all unit aliases, adding dummies if necessary so that $\ell \geq 2$, put

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U is $(u_1 - 1)^2 + \sum_{i<\ell} (u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus the final literal polynomial is

$$\boxed{P = E + \sum C + \sum Q + U.} \tag{14}$$

It is nonnegative over the reals. All monomials have total degree at most four. The full coefficient files are independently checked to have exact degree four, every nonzero coefficient equal to $+1$ or -1, constant coefficient one, and every stated variable present. The new square-alias transformation below preserves this energy identity and revises a selected equality penalty with its exact zero set intact.

4.4 Forward map, inverse map, and uniqueness

Given a source tuple, evaluate the monic affine, constant and scale constructors, assign each ordinary alias its logical value and every unit alias the value one. Assign equality dummies the value of their class. All equality and unit penalties then vanish.

Root each internal tree at its first listed vertex. For a tree edge with child v, let $\tau_v = \pm 1$ be its incidence sign at the child, and set

$$w_v = -\tau_v \sum_{t \text{ in the subtree of } v} b_t.$$

This is an explicit integral polynomial in the source coordinates. Eliminating leaves proves that all nonroot divergences vanish. Formula (13) gives root divergence T_r in each row block and $-\sum_r T_r$ in the hub block. Thus the canonical lift satisfies, universally,

$$P(L(x)) = \frac{1}{2} \left(\sum_r R_r(x)^2 + \left(\sum_r \delta_r R_r(x) \right)^2 \right). \tag{15}$$

where δ_r is the certified sign of the normalized row descending from the original residual R_r. Constructor rows pull back to zero. In particular every source zero has an integer lift. The affine-definition rows also vanish identically under their canonical substitution, so (15) composes

unchanged through affine refactorization. One must not replace (15) by a bare sum of the original squares.

Conversely, if $P(z) = 0$, nonnegativity in (14) forces every equality penalty and U to vanish. Hence all ordinary aliases in a class agree, and all unit aliases equal one. Also $E = 0$ forces every vertex divergence to vanish. Summing them over a row block cancels its internal tree flows; by the certified occurrence-boundary identity the result is precisely $T_r(y) = 0$. Since $\sigma_r = \pm 1$, all normalized rows $S_r(y)$ vanish. Triangular monic constructors then force the constant, scale and affine coordinates to their intended values. Reading the original logical coordinates yields an original source zero.

The same argument proves uniqueness: constructors, aliases, units and dummies have unique values over the original tuple, and leaf elimination uniquely determines every internal tree flow. Thus the displayed forward map and coordinate projection are inverse bijections between integer zero sets. No sign restriction on a final variable, division by a variable, or unproved boundedness assertion is used.

4.5 Square polarization removes equality dummies

Suppose a logical value currently has exactly two factor aliases a, b, and its equality penalty uses a fresh dummy d:

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

The reason for retaining this dummy can be a prescribed square a^2 inside the incidence graph: directly substituting the two-alias penalty $Q(a, b)$ may collide with its degree-four diagonal contribution. The new operation first changes selected prescribed squares into products of same-logical aliases, $a^2 \mapsto ab$ (or $b^2 \mapsto ab$). It then replaces $C(a, b, d)$ by $Q(a, b)$ and deletes the dummy coordinate d.

Theorem 4.1. *Make any finite selection of such same-logical prescribed-square changes, and replace each affected two-alias equality cycle by Q. If the full collected final polynomial has coefficient height one, it is a height-one quartic with an integer zero set bijective to the original arithmetic source. Each deleted equality dummy reduces its dimension by one.*

Proof. Changing a prescribed edge's label from a^2 to ab leaves its two incidence endpoints and orientation unchanged. The graph energy therefore remains exactly $\frac{1}{2}\sum_v d_v^2$, with its divergences evaluated using the modified labels. It is still nonnegative, and all labels still have degree at most two. Also

$$Q(a, b) = (a - b)^2(a^2 + ab + b^2) \geq 0,$$

and $Q(a, b) = 0$ if and only if $a = b$: the second factor equals $(a + b/2)^2 + 3b^2/4$ and vanishes only at $a = b = 0$.

At any zero of the final energy, every nonnegative equality term is zero, so same-logical aliases agree and units equal one. Consequently every polarized label ab has exactly the value previously specified by a^2. The occurrence-boundary identity is unchanged after logical substitution. Vanishing vertex divergences therefore give all normalized source rows, and monic reconstruction gives an actual arithmetic source zero.

Conversely, for an original source zero assign both a and b their common logical value. All polarized edge labels then have their old values, all new Q terms vanish, and the same unique subtree flow assignment makes every vertex divergence zero. Deleted dummies have unique old value $d = a = b$, so extension and deletion are inverse on the zero sets. This also proves that the source-polynomial lift identity (15) survives universally, not only at source zeros.

The degree is at most four by construction. Exact degree four and coefficient height one are finite properties of the entire collected polynomial and are explicitly checked. They do not follow merely from the simplicity of the incidence graph after monomial substitution. The operation is accepted only after this full collection check, including all equality and unit terms. Thus possible collisions do not invalidate the claim. □

The theorem makes no assertion that an arbitrary polarization is safe for height one, nor that a chosen finite sequence is optimal. It separates the unconditional witness proof from the coefficient test selecting a particular literal endpoint.

4.6 Selected outputs and matched source controls

Endpoint	Variables	Terms	Height	Constant
`minimum528`	528	20816	1	1
`compact534`	534	18543	1	1
`balanced580`	580	12001	1	1
`sparse620`	620	8318	1	1
`sparser717`	717	7176	1	1

All five have exact degree four, unrestricted integer coordinates and every declared coordinate present. Their complete coefficient arrays and literal polynomial text are supplied. The first two use the specifically identified `exposed247` source with 247 variables, 179 rows, 4889 terms and height 170496. This is the stage after the first three affine projections, but before the later height-reducing residual basis changes. It is not the literal `balanced247` equation.

The fixed selection of 24 affine square rows expands this source to 271 variables and 203 rows; coefficient normalization gives 345 variables and 277 rows. The two final graphs use 20 and 24 hub vertices. Polarization then removes ten and eleven equality dummies, respectively. These are arithmetic-source and compiler improvements together. The 534-variable, 18543-term point improves both dimension and support over the Round 18 571-variable, 19833-term point. The 528-variable point saves a further six coordinates but has more support.

To isolate polarization alone, the replay also rebuilds the exact four Round 18 graph certificates before making the new change. It checks their historical hashes and holds the arithmetic source and graph parameters fixed:

Matched source	Before variables/terms	After variables/terms
Round 18 compact	571/19833	559/19809
Round 18 balanced	593/12027	580/12001
Round 18 sparse	646/8368	620/8318
Round 18 sparser	742/7226	717/7176

Thus each matched-source transformation improves both displayed measures. The 559-variable control is reproducible but is not selected as a principal endpoint because the new-source 534-variable point has fewer variables and fewer terms. The selected 580-variable output retains the old 263-variable arithmetic source; the last two retain the old 285-variable sparse source. No new refutation calibration is needed for any of these choices.

5 Polynomial centers and eleven integer equations

5.1 Complete polynomial definitions

The independent coordinates are exactly

$$(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}.$$

The six fixed ports are the authentic source34 numerals specified in Section 2; none is another variable. Define the following ordinary polynomial abbreviations:

$$J = 6j + 1, \qquad q = \beta J + 1, \qquad W = 12w + 4,$$
$$X = qW, \qquad Y = v, \qquad E = XY,$$
$$a = E + Y, \qquad A = a + 2, \qquad H = 4a + 3,$$
$$\Delta = a^2 + H = A^2 - 1,$$

$$
\begin{aligned}
F &= (K_0 + W)C - z(q-1),\\
R &= [q(q-F) - C](q^2-1) + (M_C + qM_F)J,\\
k &= R + 1 + hE, \qquad \eta = c - Yk,\\
D_{\mathrm{err}} &= (\mathrm{cone_A}F - q)^2 + (\mathrm{cone_B}C - q)^2,\\
B_0 &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2,\\
\mathcal{B} &= Y^2B_0 - k^2q^8, \qquad D = \mathcal{B} - 4,\\
\ell &= XY^2, \qquad D_t = 1 + \ell(\ell+1)k^2,\\
D_g &= 1 + DD_ts^2, \qquad P = D_tD_g.
\end{aligned}
$$

The cofactor, auxiliary norm and carrier registers are

$$
\begin{aligned}
D_m &= 1 + \Delta c^2, & B &= rAD_m, & U &= rc^2D_m,\\
I_2 &= c^4B = Ac^2U, & D_f &= 1 + \Delta I_2, & Q &= \Delta^2 I_2,\\
\delta &= Q - 1, & V &= D_f(c - R + 4AUT) - c, & C_y &= QV^2 - 1,\\
K &= X(3c - 2X) + 2, & M &= 2q^3, & D_c &= HM,\\
S &= srHM^5,
\end{aligned}
$$

$$
z_c = D_cT + MK + H(v - q^3), \qquad A_c = (z_c^2 + 3D_c^2)(z_c^3 - 19D_c^3).
$$

The following inherited Round 18 centers fix the comparison boundary:

$$
\begin{aligned}
\Psi_{316} &= [(8\Delta + 4)U - rc^4]V, & L_{316} &= 8,\\
\Psi_{308} &= \Psi_{316} + 2U^3T(c^2 - 2), & L_{308} &= 8,\\
\Psi_{304} &= \Psi_{316} + U^2T[2U(c^2 - 2) + 16A^3], & L_{304} &= 8,\\
\Psi_{300} &= [(32\Delta + 16)U - 4rc^4]V\\
&\quad + U^2T[8U(c^2 - 2) + 64A^3 + 3rc^6], & L_{300} &= 32.
\end{aligned}
$$

Every center is an integer polynomial independent of n. For any selected even weight L and integer center Ψ, put

$$
\begin{aligned}
Z &= S(n + \Psi) - A_c,\\
p_a &= \delta Z^2, \qquad p_b = \delta S^2P, \qquad p_c = L^2S^2BD_fC_y,\\
\mathcal{N}_{L,\Psi} &= (p_a + p_b - p_c)^2 - 4p_ap_b.
\end{aligned}
\tag{16}
$$

The equation is $\mathcal{N}_{L,\Psi} = 0$ over $\mathbb{Z}^{11}$. The displayed centers are retained comparison controls. This round proves a sharp degree floor within this fixed family; it does not claim a new lower-degree scalar consistency equation. No division, radical or variable exponent is an operation of this polynomial. The literal graph shares the identities

$$
P = D_t + (\mathcal{B} - 4)(D_ts)^2,
$$

$$
\mathcal{B} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^3)^2], \qquad A^2 = \Delta + 1.
$$

Sharing these registers changes graph cost, not the polynomial.

5.2 An even weight and integral center lemma

Theorem 5.1. *For every fixed nonzero even L and every integer polynomial Ψ in the ten coordinates other than n, the projection of the integer zero set of* (16) *onto those ten coordinates equals that of the retained Round 17 free-scale equation. The latter uses $L = 2$ and $\Psi = (2\Delta + 1)UV$.*

Proof. We first record the unconditional arithmetic facts from Sections 2–4 of the retained `LINEAR_COFACTOR_ROUND17_PROOF.md`. They do not assume a zero of the present norm. The authentic affine restrictions give

$$\begin{aligned} q &\equiv 0 \pmod 2, \qquad q \equiv 2 \pmod 3, \qquad |q| \geq 16, \\ X &\equiv 0 \pmod 8, \qquad X \equiv 2 \pmod 3, \qquad |X| \geq 4|q|. \end{aligned}$$

Thus $A \neq 0$, $\Delta = A^2 - 1 > 1$, $\Delta \equiv 0 \pmod 3$, H is odd and nonzero, and $D_m, D_t \geq 1$. The exact modulo-sixteen budget identity gives $D \equiv 8$ or $12 \pmod{16}$, so D is nonzero and nonsquare. Also $P \equiv 1 \pmod 4$ and $P \neq 0$. Both D_f and δ are nonzero, Q is even, and δ is odd.

The rational polynomial $\pi(t) = (t^2+3)(t^3-19)$ has no rational root. Since $D_c \neq 0$,

$$A_c = D_c^5 \pi(z_c/D_c) \neq 0.$$

If $S = 0$, the output norm is $(\delta A_c^2)^2 \neq 0$. Every new zero therefore has $S, s, r, B \neq 0$.

Put $D_a = \delta C_y$ and $C_* = BD_fD_a$. One has $D_a > 0$: if $c = 0$, then $Q = 0$ and $D_a = 1$; if $c \neq 0$ and $B < 0$, then $\delta, C_y < 0$; if $c \neq 0$ and $B > 0$, then $D_f > |c| + 1$, so $V = D_f b - c$ is nonzero for every integer b, and $\delta, C_y > 0$. Moreover $BD_f > 0$ whenever $c \neq 0$. Hence $C_* \neq 0$, and its only possible negative case is $c = 0, B < 0$.

The four-conjugate norm identity for (16), after normalizing by nonzero S and multiplying by δ, gives a rational signed sum

$$\pm L\sqrt{C_*} \ \pm \delta\sqrt{P} \in \mathbb{Q}. \tag{17}$$

If exactly one radicand is negative, its nonzero imaginary contribution cannot cancel. If both are negative, then $c = 0$, $\delta = -1$ and $C_* = B$. Cancellation would force $L^2 B = P$, impossible modulo four because L is even and $P \equiv 1 \pmod 4$. Therefore $C_*, P > 0$.

Write these positive integers as squares times positive squarefree integers. Different nontrivial squarefree radicals are independent over $\mathbb{Q}$. The only remaining possible irrational cancellation would have the same nontrivial squarefree part in both terms of (17). Its first coefficient would be even, whereas its second coefficient would be odd: L is even and δ, P are odd. Cancellation is impossible. Thus C_* and P are positive integer squares.

The cofactor coprimalities are unchanged:

$$\gcd(B, D_f) = \gcd(B, D_a) = \gcd(D_f, D_a) = 1.$$

Indeed $D_f \equiv 1 \pmod B$ and $Q \equiv 0 \pmod B$, giving $D_a \equiv 1 \pmod B$. Modulo D_f, one has

$$Q \equiv -\Delta, \quad \delta \equiv -A^2, \quad V \equiv -c, \quad C_y \equiv -D_m, \quad D_a \equiv A^2 D_m.$$

Since $B = rAD_m$, while $D_f \equiv 1$ modulo both A and D_m, these imply the last coprimality. Therefore $|B|, |D_f|, D_a$ are squares. As $D_a > 0$, B, D_f have the same sign. But $D_f \equiv 1 \pmod 3$ excludes a negative square, so there are positive integers u, f, e with

$$B = u^2, \qquad D_f = f^2, \qquad D_a = e^2. \tag{18}$$

Similarly $\gcd(D_t, D_g) = 1$ and $D_t \geq 1$ give $D_t = \tau^2$, $D_g = g^2$ with $\tau, g > 0$. Since $s \neq 0$, the guard forces $\mathcal{B} \geq 4$, strengthened by its modulo-sixteen class to $\mathcal{B} \geq 12$. It follows that $k, Y \neq 0$, $B_0 > 0$, and

$$Y^2 B_0 > k^2 q^8, \qquad B_0 \leq k^2 q^2, \qquad |Y| > |q|^3.$$

The centered strip from $B_0 > 0$ then excludes $c = 0$.

Define proof-only integers $I = c^2 u$ and $T_{\mathrm{old}} = uT$. By (18),

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad V = f^2(c - R + 4IT_{\mathrm{old}}) - c.$$

The rational number $y = e/\delta$ satisfies

$$QV^2 - (Q-1)y^2 = 1. \tag{19}$$

It is integral by adjacent-coefficient descent. If b is its reduced denominator, then $b^2 \mid Q-1$. After choosing signs, for $|V| > 1$ the unimodular transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves (19) and the reduced denominator, with $0 < |V'| < |V|$. The first coordinate is integral because $(Q-1)y$ is integral; denominator preservation also follows from $\gcd(2Q-1, b) = 1$. For $V > 1$ and $Q > 1$, the norm gives

$$V < y < \frac{(2Q-1)V}{2(Q-1)}.$$

The squared upper bound minus y^2, multiplied by $4(Q-1)^2$, is $V^2 + 4(Q-1) > 0$. Consequently $0 < V' < V$. Descent reaches $|V| = 1$, where $|y| = 1$, forcing $b = 1$. The formal $c = 0$ case would directly give $y = -1$, although it is already excluded here.

The two normalized roots of the original four-conjugate equation are now integers $Lufy$ and τg. Hence Z/S is integral. Since Ψ is integral, $S \mid A_c$. No size or sign condition on Ψ was used. Keep the ten outer coordinates of the new zero and set

$$n_{\mathrm{old}} = A_c/S + 2ufy + \tau g - (2\Delta + 1)UV.$$

This is an integer and makes one old conjugate factor vanish, producing an actual Round 17 zero. Conversely an old zero supplies these same integral roots and divisibility by the retained theorem. Keeping its ten outer coordinates, set

$$n_{\mathrm{new}} = A_c/S + Lufy + \tau g - \Psi.$$

One new factor vanishes. These constructions prove equality of projected zero sets. They need not be mutually inverse on n, since the norm can have several sign branches. For a fixed L, changing only the center is separately a literal bijective integral shear of n. □

5.3 The complete inherited calibration chain

Theorem 5.1 reduces both directions to an actual old zero, not merely a list of necessary Pell properties. Thus its inherited scale, sign, rank and syntactic interfaces remain exactly applicable. For clarity, the retained source proof proceeds as follows.

Carrier divisibility first gives $M \mid H(v - q^3)$, even Y, and

$$H \mid q^{15}K_q^5, \qquad K_q = X(X + 2ac) - 1.$$

The strong and auxiliary pairs, together with $c^2 \mid I$, recover the main Pell root by absolute folding. With $D_t = \tau^2$, the centered strip, $|Y| > |q|^3$ and the authentic packing inequalities

$$|q|(|q|^2 - 1) < R < |q|^4 - 2,$$

the signed rank theorem yields

$$c = \mathrm{sgn}(c)\psi_A(R), \qquad R = 2t + 1, \qquad |H| > |X|(|X| - 3)^t.$$

The powered projection at the actual large base forces $\mathrm{sgn}(c)X = 2^R$. Reduction modulo three fixes $c > 0$ and $X = 2^R$; the old strip fixes $k, Y > 0$. Exact $H \mid K$ restores $Y = (2v_{\mathrm{old}} + 1)q^3$. The uniform half-binomial theorem then restores the positive native interval, and fresh odd strong

witnesses produce an actual signed Q41 source zero. Only that restored zero is passed to the retained source34/native/compiler/finite-refutation calibration.

Conversely, canonical source witnesses give $\mathcal{B} > 64$. After choosing the guard pair s, g, the old proof chooses one strong pair I, f with $4sAdc^2 \mid I$, where $D_m = d^2$, and sets

$$u = I/c^2, \qquad t_{\text{strong}} = I/(Adc^2), \qquad r = At_{\text{strong}}^2.$$

That pair is kept fixed. The supported-modulus theorem interpolates T_{old} modulo usr using the same pair, while an intersective root of π modulo sr makes A_c/S integral. There is no reselection of the strong pair after a residue request. The present change replaces only its final integer n by n_{new} above and introduces no new interpolation or choice-order assumption. Full inherited proofs and their source identities are retained separately; they are not promoted here to a newly checked formal-kernel endpoint.

5.4 A sharp degree floor for this fixed frame

The following negative result explains why further polynomial-center search alone cannot lower the degree below the Round 18 value. It is a theorem about a precisely fixed polynomial family, not a general Diophantine complexity bound.

Theorem 5.2. *Keep all the scalar registers defined above, the authentic six coefficients, and the eleven independent coordinates fixed. For any nonzero fixed integer L and any integer polynomial Ψ in the ten coordinates other than n,*

$$\deg \mathcal{N}_{L,\Psi} \geq 300.$$

The center Ψ_{300} with $L = 32$ attains degree 300. Thus 300 is the exact minimum in this fixed-frame, constant-weight, integer-center family. The degree assertion does not require L to be even; the logical zero-set theorem above does.

Proof. All ports are constants of degree zero and $\beta \neq 0$. A subscript zero in this proof denotes the complete highest homogeneous component, not an arbitrarily chosen leading monomial. Set

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad a_0 = X_0 v, \quad F_0 = 12wC - zq_0,$$

$$k_0 = R_0 = -q_0^3 F_0, \qquad \ell_0 = X_0 v^2.$$

These are nonzero polynomials of degrees $1, 2, 3, 2, 5, 4$, respectively. Exact substitution in the literal registers gives

Register	Degree	Highest homogeneous component
A	3	a_0
Δ	6	a_0^2
D_m	8	$a_0^2 c^2$
B	12	$r a_0^3 c^2$
U	11	$r a_0^2 c^4$
D_f	22	$r a_0^5 c^6$
Q, δ	28	$q_* = r a_0^7 c^6$
V	37	$v_* = 4r^2 a_0^8 c^{10} T$
C_y	102	$q_* v_*^2$
D_t	18	$\ell_0^2 k_0^2$
$\mathcal{B}$	18	$-k_0^2 q_0^8$
P	56	$-p_*^2$, where $p_* = q_0^4 \ell_0^2 k_0^3 s$
S	20	$s_* = 128 s r a_0 q_0^{15}$

In particular the highest form of P is a negative square. To check this sign directly, the terms $Y^2k^2(q^2 - D_{\mathrm{err}})$ and $-4Y^2q^2\eta^2$ in $\mathcal{B}$ have degree at most 16, whereas $-k^2q^8$ has degree 18. Since

$$P = D_t + (\mathcal{B} - 4)D_t^2 s^2,$$

its degree-56 part is exactly $-p_*^2$. This is the invariant that prevents a degree-300 cancellation.

The carrier has degree 35. Indeed its leading factors are

$$(z_c)_0 = 8a_0q_0^3T - 4q_0^3X_0^2, \qquad (A_c)_0 = ((z_c)_0)^5.$$

The two summands of $(z_c)_0$ cannot cancel because one contains T and the other does not.

Write $d = \deg \Psi$, treating the zero polynomial as degree $-\infty$. If $d < 54$, then $\deg Z \le 73$ and $\deg p_a \le 174$, whereas $\deg p_b = 124$ and $\deg p_c = 176$. Thus the inner expression $H_* = p_a + p_b - p_c$ has degree 176. The cross term has degree at most 298, so $\deg \mathcal{N} = 352$.

If $d > 54$, then $\deg Z = 20 + d$ and $\deg p_a = 68 + 2d > 176$. It dominates both other terms of H_*. Consequently

$$\deg \mathcal{N} = 136 + 4d > 352,$$

because $\deg(4p_ap_b) = 192 + 2d < 136 + 4d$.

It remains to consider $d = 54$. Let ψ_0 be the nonzero degree-54 part of Ψ. The leading forms of p_a and p_b are respectively $q_*s_*^2\psi_0^2$ and $-q_*s_*^2p_*^2$, of degrees 176 and 124. Therefore the degree-300 part of the negative cross term is

$$(-4p_ap_b)_{300} = (2q_*s_*^2\psi_0p_*)^2 \neq 0. \tag{20}$$

Put $h = \deg H_*$. If $h < 150$, this cross term proves degree 300. If $h > 150$, H_*^2 proves degree $2h > 300$. If $h = 150$, the degree-300 part is the sum of the square of the highest form of H_* and the nonzero square in (20). A sum of real polynomial squares with one nonzero summand is not the zero polynomial. Thus in all cases $\deg \mathcal{N} = \max(2h, 300) \ge 300$. Attainment is verified next. □

This theorem does not cover different carriers, changed auxiliary norms, polynomial weights, coordinate changes in the other ten variables, or a different encoding of refutation. It does not imply a global minimum even among eleven-variable equations. Within its stated family, however, an unsuccessful center search has a mathematical explanation rather than merely an exhausted search budget.

5.5 The attaining center and exact highest degree

For completeness, the attained bound is checked at the actual coefficients, not inferred from a syntactic degree estimate. Work first in the polynomial ring with independent symbols A, c, r, T, R and weights $(3, 1, 1, 1, 5)$. Keep the exact identities

$$\Delta = A^2 - 1, \quad U = rc^2(1 + \Delta c^2), \quad B = rA(1 + \Delta c^2), \quad BD_f = B + \Delta A^2U^2.$$

Define

$$\begin{aligned} \lambda &= (8\Delta + 4)U - rc^4, & \Psi_0 &= \lambda V, \\ h_* &= 2U^3T(c^2 - 2), & b_* &= 16A^3U^2T, \\ e_* &= 3rc^6U^2T. \end{aligned}$$

The attaining center is $\Psi_{300} = 4(\Psi_0 + h_* + b_*) + e_*$. Its squared approximation error $D_{300} = \Psi_{300}^2 - 32^2BD_fV^2$ follows from the exact identities

$$\begin{aligned} D_{316} &= [8Ur(2c^2 - c^4) + r^2c^8 - 64B]V^2, \\ D_{308} &= D_{316} + 2\Psi_0h_* + h_*^2, \\ D_{304} &= D_{308} + 2(\Psi_0 + h_*)b_* + b_*^2, \\ D_{300} &= 16D_{304} + 8(\Psi_0 + h_* + b_*)e_* + e_*^2. \end{aligned}$$

These are literal polynomial identities obtained by squaring and using $BD_f = B + \Delta A^2 U^2$. Expand with $U = r(c^4A^2 + c^2 - c^4)$ and $V = D_f(c - R + 4AUT) - c$. The highest weighted part of D_{300} is

$$4096A^{16}c^{26}r^6T^2,$$

of weight 82; every remaining term has smaller weight. The supplied exact coefficient expansion verifies the cancellation of every term of greater weight, not only a numerical sample.

Set $E_{300} = \delta\Psi_{300}^2 - 32^2BD_fC_y$. Since $C_y = QV^2 - 1$ and $\delta = Q - 1$,

$$E_{300} = \delta D_{300} - 32^2BD_f(V^2 - 1).$$

The second summand has weight 108; the first has highest part $4096A^{23}c^{32}r^7T^2$ of weight 110. Put $W_* = Sn - A_c$ of degree 35. The exact norm cut is

$$p_a + p_b - p_c = \delta W_*^2 + 2\delta S\Psi_{300}W_* + \delta S^2P + S^2E_{300}.$$

The first three terms have degrees at most 98, 137 and 124, respectively, whereas the last has exact degree 150. Its square thus has degree 300. A nonzero T^4 term in that degree, before expanding a_0, q_0, is

$$2^{52}s^4r^{18}c^{64}T^4a_0^{50}q_0^{60}.$$

The leading cross term has T-degree only two, so it cannot cancel this monomial. This independently establishes attainment; it agrees with the positive-square argument of Theorem 5.2.

5.6 What this scalar iteration establishes

The unchanged attaining control has eleven unrestricted integer variables, exact degree 300 and 148 live binary operations. No scalar degree, dimension or operation-count improvement is claimed in this iteration. The new result is the exact degree minimum within the fixed constant-weight, integer-center family of Theorem 5.2.

Going below that bound requires changing another part of the arithmetic frame and proving its witness implications afresh. In particular, a carrier congruence for a product cannot be cancelled using a coprimality statement about a different pair of factors. The existing scale guard is retained; no unproved guard deletion is used by an accepted endpoint.

The finite scalar checker binds the attaining graph to its authentic coefficient payloads and verifies the complete highest homogeneous forms used in the degree theorem. This exact algebra is distinct from the retained unbounded Pell, interpolation and refutation calibration. Neither successful replay nor the constrained degree minimum asserts consistency or any global minimum among eleven-variable Diophantine equations.

6 Comparison with Round 18 and the newest inputs

6.1 The fixed comparison snapshot

The repository comparison snapshot is ProveIt commit

`5d22cbf6bf2a344b6abe876e757b3fdc496d4adc`,

authored at 21:34:00 UTC on 25 October 2026. The intake records bind its Git tree and archive objects. This is newer than the Round 18 comparison pin

`e0c98c629bf10d30b695884f6b1d2ab361bdf513`.

Comparing only with that older pin would omit materially better direct equations.

The seven newly considered packages are Symmetric residue budgets, Shared residue bounds, Centered residues, Common remainders and radix47, Coprime products and positive centers,

Residual bases and denominator engineering, and Symmetric remainders. They are separate sources even where names or mechanisms overlap. Immutable intake identity and a successful algebraic replay are distinguished from acceptance of an entire mathematical package. The exact inspection receipts record which literal equations were recollected and which full package checks were run.

6.2 Signed quartic comparisons

Every entry below is a degree-four equation over unrestricted integers. Support includes the constant monomial when it is nonzero.

Endpoint	Variables	Rows	Terms	Height
Round 18 dimension	258	186	5364	128446
Round 18 grouped	263	191	4141	128446
Round 18 sparse merged	285	213	3750	128446
Round 18 sparse four pools	286	212	3744	128446
Symmetric budgets principal	258	186	5319	47281
Shared bounds principal	260	186	4770	9191
Centered residues principal	258	186	5366	128446
Common remainders principal	252	182	5340	128446
Symmetric remainders principal	260	185	5099	128446
Residual bases dimension	278	188	12884	12227
Residual bases balanced	279	189	5483	7200
Residual bases sparse	279	189	5202	12662
Residual bases wider sparse	284	194	4270	10864
Residual bases widest sparse	306	216	3870	14512

The common-denominator source is the relevant new dimension baseline, not only the older 282-variable arithmetic ancestor. The new construction combines its shared normalized digits with compatible restrictions and exact coordinate or residual transformations. The shared-bounds source contributes frozen integral shears; the residual-bases source provides a useful comparison for exposing affine consequences by row changes. These ingredients are credited as inputs rather than renamed as new discoveries.

The new balanced point 244/176/5212/3968 improves every displayed varying metric of the incoming Common remainders point 252/182/5340/128446: eight fewer variables, six fewer rows, 128 fewer monomials, and lower height. It also improves every displayed varying metric of the Round 18 dimension point 258/186/5364/128446: fourteen fewer variables, ten fewer rows, 152 fewer monomials and lower height.

The new sparse point has 250 variables, 182 rows, 4770 monomials and height 6709. Relative to the Shared bounds principal, this saves ten variables and four rows, keeps the same support and lowers height. It does not dominate the Round 18 grouped point: its support is larger than 4141. The new balanced point similarly has more support than both Round 18 sparse points. Those sparse alternatives remain meaningful.

The 243-variable endpoint improves dimension and rows further, at the substantial cost of 19517 monomials and height 7680. The 246-variable alternative has 21537 terms but lower height 6144. The low-height 247-variable point reduces height to 3840 but raises support to 10422. None is labeled a simultaneous improvement over the balanced point. These are exact finite tradeoffs, not competing claims of an undefined overall simplicity record.

6.3 Height one comparisons on fixed sources

All entries in this comparison have degree four, unrestricted variables, coefficient height one and constant coefficient one. The four Round 18 variable/support pairs are

$$(571, 19833), \quad (593, 12027), \quad (646, 8368), \quad (742, 7226).$$

The new matched-source results in Section 4 hold the original arithmetic source and incidence construction fixed before polarization. They therefore isolate an equality-encoding improvement from any gain due to a new smaller arithmetic source. An endpoint compiled from a newly reduced signed source is labeled separately. There is no implication that a smaller arithmetic source must produce a smaller height-one expansion: source monomial incidence and alias multiplicity also matter.

6.4 Natural quartics are a separate domain

The following measurements are relevant context, not unrestricted-integer results. All coordinates range over $\mathbb{N} = \{0, 1, 2, \ldots\}$.

Endpoint	Variables	Rows	Terms	Height
Round 18 minimum dimension	221	171	13935	118464197
Round 18 balanced	222	172	9917	83954634
Round 18 intermediate	223	173	7202	83954472
Round 18 sparse	224	174	6527	83953869
Coprime products principal	221	172	9536	83954186
Coprime products sparse	222	173	6821	83954028
Centered residues balanced	222	172	9522	83954186
Centered residues sparse	223	173	6858	83953724
Residual bases balanced	229	179	5607	65892965

For example, the incoming Coprime products 221-variable point has fewer terms and lower height than the Round 18 221-variable point, but has one more residual. Calling this an improvement in every displayed metric would be incorrect. Its 222-variable sparse companion and the Round 18 224-variable sparse point also trade dimension against support and height. No new natural-domain theorem is silently inferred from a signed endpoint.

6.5 Few variable circuits and coefficient architectures

The Round 18 eleven-integer degree/cost pairs were

$$(316, 133), \quad (308, 140), \quad (304, 144), \quad (300, 148).$$

They use the same authentic source34 six-port tuple as the retained eleven-integer control, so degree and live graph cost are directly comparable. The earlier Round 17 alternatives $(328, 130)$, $(340, 129)$, $(352, 125)$, and $(398, 121)$ remain relevant lower-cost choices. A degree reduction at higher cost does not dominate them.

The newest Residual bases package reports unrestricted-integer points with fourteen variables at degree/cost 86/112 and 94/110, and thirteen variables at 156/125. Their smaller degrees do not make them improvements in every metric over an eleven-variable equation.

The positive-domain results form another comparison class. Centered residues reports ten strictly positive variables at cost/degree 75/626, 76/464 and 77/460. Coprime products reports ten-positive points 76/626, 77/464 and 80/382, fourteen-positive points 70/189 and 71/157, and a thirteen-positive point 72/433. Symmetric remainders gives ten-positive points 77/626, 78/464 and 79/460. These are reported compact metrics from the inspected packages; the direct coefficient recollection does not itself verify their unbounded positive-domain theorems.

The Common remainders package also supplies a coherent radix47 native coefficient tuple totaling 207214 raw bytes. This concerns a different native encoding. It is not a license to replace one or more ports in the source34 tuple while retaining its proof. A new coherent specialization requires the complete encoding and calibration bridge. The present scalar proof uses its own authentic unchanged six-port tuple throughout.

6.6 Limits of the comparison

The comparison universe is the named pinned artifacts and prior releases, not all published or possible Diophantine equations. The tables report achieved objects; they do not establish global dimension minima, arithmetic circuit lower bounds, coefficient-optimality, or literature-wide priority. Separate audits in this workflow are AI-assisted independent reasoning and implementations, not outside human peer review. No upstream commit, pull request or publication is part of preparing these artifacts.

7 Reproduction and evidence boundaries

7.1 Portable entry points

From the extracted release root, run

```
python3 -B verify_release.py
```

The integrated verifier authenticates the release inventory, copies the working tree to a disposable temporary directory, executes the explicit `VERIFY_PLAN.json` and checks that the sealed reference tree remains unchanged. The optional `-output` argument saves a fresh receipt outside the sealed release, for example

```
python3 -B verify_release.py -output ../new_verification.json
```

The `-integrity-only` option checks inventory rather than mathematical reproduction. Python and its standard library suffice for the mathematical checks; neither network access, a repository checkout, an external computer algebra system nor a proof assistant is required.

The selected signed equations are `signed/balanced244`, `dimension243`, `sparse250`, `dimension246` and `lowheight247`, with the latter four also below `signed/`. The additional `exposed247` and `balanced247` entries are technical source and construction checkpoints. Each directory supplies complete residuals, the complete collected polynomial and exact metrics. The frozen certificates record every row addition, shear and restoration.

The height-one catalog identifies `minimum528`, `compact534`, `balanced580`, `sparse620` and `sparser717`. Each has a complete coefficient array, ordinary literal polynomial text, a full graph certificate and its source binding. The scalar material retains the authentic degree-300 control, its numerical payloads and the precise restricted family theorem. It does not supply a new lower-degree calibrated endpoint.

The report has a separate build command:

```
sh report/build_report.sh
```

A standard TeX installation is needed only for that command. The builder uses a temporary directory, runs three passes, retains compilation logs and rejects overfull boxes or unresolved references. The principal residual appendix is generated directly from the literal JSON by `report/render_residuals.py`; its binding receipt records both source and generated-TeX hashes. PDF bytes can vary with TeX versions and metadata; the literal equation arrays are normative mathematical data.

7.2 What was independently checked

The final component receipts record the following checks in ordinary and optimized Python. Required tests use explicit exceptions, so disabling Python assertions does not disable the checks.

- The signed checker imports no producer. It authenticates the exact centered source, recognizes the deleted guard and the eleven changed common-sphere coefficients, and derives the new odd-square row from the recorded translation. It replays every inverse row or coordinate

operation, every unit restoration and every index compaction. For quadratic restorations it checks the only-linear-occurrence condition. It recollects every coefficient of all seven emitted arrays and rejects malformed coefficients, restorations, noninvertible changes and invalid nonlinear substitutions.

- The height-one checker reconstructs every coefficient from divergence squares and equality/unit identities. It checks all source pullbacks, occurrence boundaries, tree edges, aliases, signs and the universal symbolic lift at every vertex. All five selected profiles passed, including deliberate polarization and other certificate corruptions. Four matched-source replays first reproduce the original Round 18 graph hashes, then check the new polarization results. Isolated release-only normal and optimized rebuilds also passed.
- The independent scalar highest-form audit checks the exact registers at the authentic coefficients, including the complete negative-square highest part of P. Those finite identities supply the inputs to the unbounded polynomial degree argument. The separate scalar checker reconstructs all six numerical payloads, binds every live gate of the attaining control, checks its exact center-remainder certificate, and rejects both alternative operation labels at each gate. The graph rebuild is byte identical.
- A separate implementation recollected all coefficients of 44 incoming direct arrays, including intermediate and comparison endpoints, in normal and optimized modes. Their supplied expansions and measured domains, dimensions, degrees, supports and heights agreed. This does not certify all unbounded theorems of the corresponding incoming packages.

These are component execution results, not an inference from merely listing commands. The integrated release receipt records which sealed-plan steps actually ran. The mathematical audits separately checked the normalized sphere and odd-radix witness arguments, the polarization bijection and the restricted scalar degree theorem.

7.3 The retained logical boundary

The common-remainder arithmetic input and its portable ordinary source closure are retained under `prior_direct_closure`. The earlier signed sources used by the height-one comparisons are bound separately under `round18_closure`. They preserve the Pell, CRT, factorial, orientation, strict-range and finite-calculus arguments and their named C222/ZF/ZFC interfaces. The scalar branch retains its separate source34 numerical compiler, native calibration, signed-rank, carrier and interpolation route. No coefficient tuple or domain is silently exchanged between branches.

Authentication establishes file identity, not mathematical truth. Exact coefficient replay proves finite algebra, not an unbounded classification theorem. The ordinary witness proofs compose with the explicitly identified inherited calibration; a hash, an included Lean interface file or a successful Python run does not rebuild a transitive proof-assistant development. No new end-to-end kernel check or encoded object-language ZFC derivation is claimed. Nothing in these checks decides whether ZFC is consistent or whether any calibrated equation has a zero.

8 Further questions

8.1 Can the common budget remove another arithmetic witness

The normalized common denominator compresses all eleven intervals into one bound, but it still pays for the ratio coordinates and for the surviving front sign certificates. Can one remove a ratio coordinate without introducing a cubic residual, or use an existing exact power or ceiling identity to obtain a monic quadratic restoration? Any proposed change must preserve the orientation-first order and construct complete witnesses, not merely derive a necessary inequality at a hypothetical zero.

8.2 Joint optimization of residuals and aliases

An elementary residual-row operation may reveal a monic coordinate while making coefficient height or incidence density worse. Conversely, a denser arithmetic presentation may admit more same-logical alias identifications. A useful finite problem is to optimize a fixed source jointly over certified row operations, degree-preserving eliminations and polarized alias graphs. A certificate of optimality in that constrained class would be meaningful; it would not establish a minimum over all Diophantine equations.

8.3 Move beyond the exhausted fixed scalar frame

Theorem 5.2 proves that changing only a constant weight and integral polynomial center cannot lower degree below 300 in the fixed frame. Further degree reduction therefore requires a different budget, carrier, auxiliary norm or change in the other coordinates. Any shortened sign or divisibility argument must prove the required coprimality before cancelling a factor; a congruence for a product does not generally imply the corresponding congruence for either factor.

8.4 A checked connection from arrays to refutation

The finite source restorations and energy identities are suitable targets for proof-assistant formalization. A useful next milestone would bind an actual emitted coefficient array to the existing arithmetic relation, then rebuild the full calibration chain to syntactic ZFC refutation. A hash manifest, a source file naming a theorem, or a finite optimized-interpreter run is not a substitute for that transitive kernel check.

A Every residual of the balanced principal equation

The following is the complete finite definition of the principal polynomial. Its independent coordinates are $x_0, \ldots, x_{243} \in \mathbb{Z}$, and

$$P(x_0, \ldots, x_{243}) = \sum_{i=0}^{175} R_i(x)^2.$$

Indices are zero based, exactly as in the literal JSON array. No register or additional quantified variable is implicit in this appendix. The adjacent collected coefficient file specifies the same polynomial.

$$\begin{aligned}
R_0 &= -x_3^2 + x_4 \\
R_1 &= x_0 - x_0x_2 - x_2 + x_9 \\
R_2 &= -1 + x_1 - x_2x_7 - x_3 \\
R_3 &= -1 + x_1 - x_8x_9 \\
R_4 &= 1 + x_4 - x_5x_{20} + x_{18} \\
R_5 &= -1 + x_{10} - x_{10}^2 - 2x_{10}x_{11} - x_{11} - x_{11}^2 \\
&\quad + x_{21} \\
R_6 &= -5 - 20x_{10} - 16x_{10}^2 - 16x_{10}x_{11} - 16x_{10}x_{21} - 10x_{11} \\
&\quad - 4x_{11}^2 - 8x_{11}x_{21} - 10x_{21} - 4x_{21}^2 + x_{22} \\
R_7 &= -2 - x_{10} - x_{10}^2 - 4x_{10}x_{22} - 4x_{22} - 4x_{22}^2 \\
&\quad + x_{23} \\
R_8 &= 1 + 2x_{23} - 4x_{23}^2 + x_{24} \\
R_9 &= -2 - x_{11}^2 - 2x_{11}x_{12} - x_{12} - x_{12}^2 + x_{25} \\
R_{10} &= -9 + 7x_{10} - x_{10}^2 - 4x_{10}x_{11} + 4x_{10}x_{12} - 4x_{10}x_{25} \\
&\quad + 12x_{11} - 4x_{11}^2 + 8x_{11}x_{12} - 8x_{11}x_{25} - 15x_{12} - 4x_{12}^2 \\
&\quad + 8x_{12}x_{25} + 12x_{25} - 4x_{25}^2 + x_{26} - x_{32} \\
R_{11} &= -1 + x_{10} - x_{10}^2 - 2x_{10}x_{12} - x_{11} - x_{12}^2 \\
&\quad + x_{27} \\
R_{12} &= 11 + 6x_{10} - 40x_{10}^2 + 6x_{10}x_{11} - 32x_{10}x_{12} - 32x_{10}x_{21} \\
&\quad + 4x_{10}x_{22} - 16x_{10}x_{27} + 8x_{11}^2 - 8x_{11}x_{12} - 4x_{11}x_{27} + 15x_{12} \\
&\quad - 16x_{12}x_{21} - 9x_{21} - 4x_{21}^2 - 8x_{21}x_{27} + 4x_{22} + 4x_{22}^2 \\
&\quad - x_{23} - x_{26} + 8x_{27} + x_{28} + 2x_{32} - x_{33} \\
R_{13} &= -36x_{12}^2 + 24x_{12}x_{28} - 4x_{28}^2 + x_{29} \\
R_{14} &= -6 - 8x_{10} - 4x_{10}^2 - 8x_{10}x_{11} - 8x_{10}x_{12} + 8x_{10}x_{25} \\
&\quad + 8x_{10}x_{27} - 6x_{11} - 4x_{11}^2 - 8x_{11}x_{12} + 8x_{11}x_{25} + 8x_{11}x_{27} \\
&\quad - 5x_{12} - 4x_{12}^2 + 8x_{12}x_{25} + 8x_{12}x_{27} - x_{21} + 8x_{25} \\
&\quad - 4x_{25}^2 - 8x_{25}x_{27} + 6x_{27} - 4x_{27}^2 + x_{30} \\
R_{15} &= 1 - 4x_{10} + 2x_{11} - 4x_{11}^2 - 8x_{11}x_{12} - 8x_{11}x_{30} \\
&\quad + 4x_{12} - 4x_{12}^2 - 8x_{12}x_{30} - 2x_{21} + 4x_{30} - 4x_{30}^2 \\
&\quad + x_{31} \\
R_{16} &= 29 + 8x_{10} + 10x_{10}^2 + 10x_{10}x_{11} + 4x_{10}x_{12} - 20x_{10}x_{25} \\
&\quad - 16x_{10}x_{27} - 21x_{11} - 13x_{11}^2 - 32x_{11}x_{12} - 40x_{11}x_{25} - 16x_{11}x_{27} \\
&\quad - 11x_{12} - 16x_{12}^2 - 40x_{12}x_{25} - 16x_{12}x_{27} - x_{21} - 40x_{25} \\
&\quad + 4x_{25}^2 + 16x_{25}x_{27} - 12x_{27} + 8x_{27}^2 - 2x_{30} + x_{32} \\
R_{17} &= 6 - 24x_{10} - 8x_{10}^2 + 8x_{10}x_{11} - 16x_{10}x_{12} - 16x_{10}x_{27} \\
&\quad + 13x_{11} - x_{11}^2 + 8x_{11}x_{12} + 4x_{11}x_{27} - 15x_{12} - 8x_{12}^2 \\
&\quad - 16x_{12}x_{27} - 16x_{27} - 4x_{27}^2 - x_{32} + x_{33} \\
R_{18} &= 2 - 28x_{12}^2 - 24x_{12}x_{28} - 32x_{12}x_{33} + 4x_{28}^2 - x_{29} \\
&\quad - 4x_{33}^2 + x_{34} \\
R_{19} &= -3 + 5x_{10} - 4x_{10}x_{11} + 4x_{10}x_{12} + 4x_{10}x_{22} - 4x_{10}x_{25}
\end{aligned}$$

$$
\begin{aligned}
&- 35x_{11} - 32x_{11}^2 + 24x_{11}x_{12} + 24x_{11}x_{21} + 16x_{11}x_{25} - 8x_{12} \\
&+ 4x_{12}^2 + 8x_{12}x_{25} + 16x_{21} - 4x_{21}^2 - 8x_{21}x_{25} + 4x_{22} \\
&+ 4x_{22}^2 - x_{23} + 18x_{25} - 8x_{25}^2 + x_{26} - x_{27} \\
&- x_{32} + x_{35} \\
R_{20} = &-3 - 6x_{10} + 2x_{11} - 2x_{12} - 2x_{27} - 4x_{35} \\
&- 4x_{35}^2 + x_{36} \\
R_{21} = &16 + 30x_{10} + 14x_{10}^2 - 2x_{10}x_{11} - 24x_{10}x_{12} - 16x_{10}x_{21} \\
&+ 60x_{10}x_{22} - 16x_{10}x_{30} + 19x_{11} - 6x_{11}x_{12} - 24x_{11}x_{21} + 32x_{11}x_{22} \\
&- 12x_{12} + 6x_{12}^2 - 12x_{12}x_{21} + 12x_{12}x_{30} + 3x_{21} - 8x_{21}^2 \\
&+ 32x_{21}x_{22} - 8x_{21}x_{30} + 15x_{22} - 4x_{22}^2 - x_{25} - 2x_{27} \\
&+ 4x_{30}^2 + x_{37} + 8x_{40} - 8x_{40}^2 + 2x_{41} - x_{46} \\
R_{22} = &1 + 2x_{37} - 4x_{37}^2 + x_{38} \\
R_{23} = &-x_{11}^2 - 4x_{11}x_{38} - 4x_{38}^2 + x_{39} \\
R_{24} = &-7 + 9x_{10} - 9x_{10}^2 + 22x_{10}x_{11} + 16x_{10}x_{12} - 32x_{10}x_{21} \\
&+ 16x_{10}x_{27} - 6x_{11} - 12x_{11}^2 + 32x_{11}x_{21} - 4x_{11}x_{27} + 14x_{12} \\
&+ 8x_{12}^2 + 16x_{12}x_{27} - 14x_{21} - 16x_{21}^2 - 4x_{25} + 17x_{27} \\
&+ 4x_{27}^2 + x_{32} - x_{33} + x_{40} + 2x_{75} \\
R_{25} = &-4 - 3x_{10} - 7x_{11} - 4x_{12} - 2x_{25} - x_{27} \\
&+ 4x_{40} - 4x_{40}^2 + x_{41} \\
R_{26} = &-6x_{10} + 2x_{11} + 2x_{12} - 4x_{12}^2 - 8x_{12}x_{41} - 2x_{27} \\
&+ 4x_{41} - 4x_{41}^2 + x_{42} \\
R_{27} = &-x_{12} - x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{43} \\
R_{28} = &-31 + 2x_{10} - x_{10}^2 - 4x_{10}x_{43} - 22x_{11} + 16x_{11}^2 \\
&- 32x_{11}x_{12} + 32x_{11}x_{25} - 42x_{12} - 20x_{12}^2 - 8x_{12}x_{25} + 24x_{12}x_{27} \\
&+ 12x_{25}^2 - 8x_{25}x_{27} + 22x_{27} - 4x_{27}^2 + 2x_{43} - 4x_{43}^2 \\
&+ x_{44} - 4x_{47} \\
R_{29} = &4x_{10} - 2x_{10}^2 - 8x_{10}x_{43} - 16x_{12}^2 - 16x_{12}x_{44} + 4x_{43} \\
&- 8x_{43}^2 - 4x_{44}^2 + x_{45} \\
R_{30} = &-16x_{10}^2 + 24x_{10}x_{12} - 16x_{10}x_{21} + 16x_{10}x_{30} + x_{12} - 9x_{12}^2 \\
&+ 12x_{12}x_{21} - 12x_{12}x_{30} - 4x_{21}^2 + 8x_{21}x_{30} - 4x_{30}^2 + x_{46} \\
R_{31} = &-1 + 6x_{11} - 4x_{11}^2 + 8x_{11}x_{12} - 8x_{11}x_{25} - 6x_{12} \\
&- 4x_{12}^2 + 8x_{12}x_{25} + 6x_{25} - 4x_{25}^2 + x_{47} \\
R_{32} = &-1 - x_{12} - x_{12}^2 - 4x_{12}x_{47} - 4x_{47}^2 + x_{48} \\
R_{33} = &1 + x_{11} - x_{11}^2 - 4x_{11}x_{48} - 4x_{48}^2 + x_{49} \\
R_{34} = &-1 - 6x_{49} - 4x_{49}^2 + x_{50} \\
R_{35} = &-1 - x_{12} - x_{12}^2 - 4x_{12}x_{50} - 4x_{50}^2 + x_{51} \\
R_{36} = &2 + x_{11} - x_{11}^2 - 4x_{11}x_{51} - 4x_{51}^2 + x_{52} \\
R_{37} = &-2x_{46} - 4x_{46}^2 - 8x_{46}x_{52} - 4x_{52}^2 + x_{53} \\
R_{38} = &-9 + 7x_{10} - x_{10}^2 - 4x_{10}x_{53} + 12x_{53} - 4x_{53}^2 \\
&+ x_{54} \\
R_{39} = &-1 - 6x_{54} - 4x_{54}^2 + x_{55} \\
R_{40} = &-x_{12} - x_{12}^2 - 4x_{12}x_{55} - 4x_{55}^2 + x_{56} \\
R_{41} = &-x_{10} - x_{10}^2 - 4x_{10}x_{56} - 4x_{56} - 4x_{56}^2 + x_{57} \\
R_{42} = &-10 - x_{11} + x_{12} - x_{25} - 14x_{57} - 4x_{57}^2 \\
&+ x_{58} \\
R_{43} = &1 - 2x_{11} + 2x_{12} - 2x_{25} + 4x_{58} - 4x_{58}^2 \\
&+ x_{59} \\
R_{44} = &-10 + x_{11} - x_{11}^2 - 4x_{11}x_{59} - 14x_{52} - 4x_{52}^2 \\
&- 4x_{59}^2 + x_{60} \\
R_{45} = &21 + 28x_{52} + 8x_{52}^2 - 2x_{60} - 4x_{60}^2 + x_{61}
\end{aligned}
$$

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\begin{aligned}
R_{46} &= 15x_{11} - 36x_{11}^2 + 12x_{11}x_{12} + 24x_{11}x_{21} - x_{12} - x_{12}^2 \\
&\quad - 4x_{12}x_{21} - 5x_{21} - 4x_{21}^2 + x_{62} \\
R_{47} &= 1 + 6x_{11} + x_{12} - x_{12}^2 - 4x_{12}x_{61} - 2x_{21} \\
&\quad - 4x_{61}^2 + 4x_{62} - 4x_{62}^2 + x_{63} \\
R_{48} &= -2 - 2x_{12} + 2x_{12}^2 + 8x_{12}x_{61} + 8x_{61}^2 - 2x_{63} \\
&\quad - 4x_{63}^2 + x_{64} \\
R_{49} &= -2 - 4x_{64} - 4x_{64}^2 + x_{65} \\
R_{50} &= -x_3x_{15} + x_{66} \\
R_{51} &= x_{14} - x_{15} - 2x_{24}x_{66} - x_{67} \\
R_{52} &= 12x_{12}x_{66} + x_{14} - x_{15} - 4x_{26}x_{66} - 2x_{29}x_{66} + 4x_{32}x_{66} \\
&\quad - x_{68} \\
R_{53} &= x_{14} - x_{15} - 2x_{31}x_{66} - 6x_{66} - x_{69} \\
R_{54} &= -12x_{12}x_{66} + x_{14} - x_{15} - 4x_{32}x_{66} - 2x_{34}x_{66} - x_{70} \\
R_{55} &= x_{14} - x_{15} - 2x_{36}x_{66} + 6x_{66} - x_{71} \\
R_{56} &= x_{14} - x_{15} - 4x_{38}x_{66} - 2x_{39}x_{66} - x_{72} \\
R_{57} &= x_{14} - x_{15} - 2x_{42}x_{66} - x_{73} \\
R_{58} &= x_{14} - x_{15} - 2x_{45}x_{66} - 4x_{66} - x_{74} \\
R_{59} &= -x_{10}x_{11} + x_{10}x_{12} + x_{11} + x_{11}x_{12} - x_{12} - x_{12}^2 \\
&\quad + x_{75} \\
R_{60} &= 1 + 2x_{13} + x_{13}^2 - x_{75}^2 + x_{76} \\
R_{61} &= -x_{76}^2 + x_{77} \\
R_{62} &= -4x_{64}x_{66} - 2x_{65}x_{66} + 2x_{66} + x_{78} \\
R_{63} &= -x_{14}^2 + 2x_{14}x_{15} + 2x_{14}x_{78} - x_{15}^2 - 2x_{15}x_{78} - x_{78}^2 \\
&\quad + x_{79} \\
R_{64} &= -1 - x_3x_{11} + x_{80} \\
R_{65} &= -x_{15}x_{80} + x_{82}x_{83} + x_{84} \\
R_{66} &= -1 + x_3 - x_3x_{10} + x_{81} \\
R_{67} &= -1 + 2x_3 + 6x_3x_{11} - 2x_3x_{21} + x_{82} \\
R_{68} &= -x_{14}x_{81} + x_{83} \\
R_{69} &= -6x_3x_{15} + x_{14}x_{80} + 12x_{15} - 8x_{15}x_{80} - 4x_{15}x_{81} - x_{15}x_{82} \\
&\quad + x_{85} \\
R_{70} &= -x_{14}x_{67} + x_{15}x_{67} + x_{86} \\
R_{71} &= -x_{68}x_{86} + x_{87} \\
R_{72} &= -x_{69}x_{87} + x_{88} \\
R_{73} &= -x_{70}x_{88} + x_{89} \\
R_{74} &= -x_{71}x_{89} + x_{90} \\
R_{75} &= -x_{72}x_{90} + x_{91} \\
R_{76} &= -x_{73}x_{91} + x_{92} \\
R_{77} &= -x_{74}x_{92} + x_{93} \\
R_{78} &= -x_{84}x_{93} + x_{94} \\
R_{79} &= -x_{85}x_{94} + x_{95} \\
R_{80} &= -x_{95}^2 + x_{96} \\
R_{81} &= -x_2 - x_2x_{18} + x_{18} + x_{97} \\
R_{82} &= -x_{16}x_{97} + x_{98} \\
R_{83} &= -2x_1^2 + 2x_1x_{14} + 2x_1x_{15} + 2x_1x_{98} + 2x_1x_{99} - x_{14}^2 \\
&\quad - 2x_{14}x_{98} - x_{15}^2 - 2x_{15}x_{99} + x_{19}x_{101} - x_{77}x_{96} - x_{79}x_{96} \\
&\quad - x_{98}^2 - x_{99}^2 \\
R_{84} &= -x_2x_{17} + x_{17} - x_{17}x_{97} + x_{99} \\
R_{85} &= -5x_{10} + 3x_{11} - 4x_{21} - 2x_{24} - 4x_{26} - 2x_{29} \\
&\quad - 2x_{31} - 2x_{34} - 2x_{36} - 4x_{38} - 2x_{39} - 2x_{42} \\
&\quad - 2x_{45} - 4x_{64} - 2x_{65} + x_{100} \\
R_{86} &= -x_{102}x_{105} + x_{105} + x_{107} \\
R_{87} &= -x_5x_{129} + x_{103}x_{228} - x_{106} - x_{107}x_{129}
\end{aligned}
$$

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\begin{aligned}
R_{88} &= x_{108}x_{229} - x_{110}x_{132} - x_{111}\\
R_{89} &= x_{112}x_{229} - x_{113}x_{132} - x_{114}\\
R_{90} &= x_{115}x_{229} - x_{116}x_{132} - x_{117}\\
R_{91} &= x_{118}x_{229} - x_{119}x_{132} - x_{120}\\
R_{92} &= x_{121}x_{230} - x_{123}x_{132} - x_{124}\\
R_{93} &= x_{125}x_{230} - x_{126}x_{132} - x_{127}\\
R_{94} &= -x_{19}x_{129} + x_{132}\\
R_{95} &= x_{128} - x_{130}x_{132} - x_{131}\\
R_{96} &= -x_{132}x_{134} + x_{133} - x_{135}\\
R_{97} &= -x_{132}x_{137} + x_{136}x_{228} - x_{138}\\
R_{98} &= -x_{132}x_{140} + x_{139}x_{229} - x_{141}\\
R_{99} &= x_{142} - 2x_{142}x_{144} + x_{145}\\
R_{100} &= -x_{142}x_{145} - x_{145} + x_{146}\\
R_{101} &= 2x_{143} + x_{143}^2 + x_{145}^2 - x_{146}^2\\
R_{102} &= -x_{143}x_{148} - x_{148} + x_{149}\\
R_{103} &= 2x_{147} + x_{147}^2 + x_{148}^2 - x_{149}^2\\
R_{104} &= x_4^2 - 2x_4x_{143} - 2x_{143} + x_{150}\\
R_{105} &= -1 - x_4x_{148} + x_6 - x_{147} - x_{148} + x_{149}\\
&\quad + x_{150}x_{151}\\
R_{106} &= 1 - 2x_5 + x_5^2 - 2x_5x_{143} + x_{152}\\
R_{107} &= -x_5x_{197} + x_{19} + x_{152}x_{153} - x_{196} + x_{198}\\
R_{108} &= -1 + 2x_{143}x_{212} - x_{154} + 2x_{212} - x_{212}^2\\
R_{109} &= -1 + x_{102} + x_{154}x_{155} - x_{196} - x_{197}x_{212} + x_{198}\\
R_{110} &= -x_{143}x_{157} - x_{157} + x_{158}\\
R_{111} &= 2x_{156} + x_{156}^2 + x_{157}^2 - x_{158}^2\\
R_{112} &= 1 - 2x_{102} + x_{102}^2 - 2x_{102}x_{143} + x_{159}\\
R_{113} &= -1 - x_{102}x_{157} + x_{103} - x_{156} + x_{158} + x_{159}x_{160}\\
R_{114} &= -1 - x_{102}x_{148} + 2x_{102}x_{161} + x_{104} - 2x_{143}x_{161} - x_{147}\\
&\quad + x_{148} + x_{149} + x_{159}x_{161} - 3x_{161}\\
R_{115} &= 1 + x_{143} - x_{143}x_{163} - x_{163} + x_{164}\\
R_{116} &= 1 + 2x_{162} + x_{162}^2 - 2x_{163} + x_{163}^2 - x_{164}^2\\
R_{117} &= -1 + x_{102} - x_{102}x_{163} + x_{108} + x_{159}x_{165} - x_{162}\\
&\quad + x_{164}\\
R_{118} &= -2x_{143} - x_{143}^2 + x_{166}\\
R_{119} &= 1 - x_{102} - x_{102}x_{147} - x_{102}x_{149} + 2x_{102}x_{167} + x_{109}\\
&\quad + x_{143}x_{149} - 2x_{143}x_{167} + x_{147} - x_{148}x_{166} + 2x_{149} + x_{159}x_{167}\\
&\quad - 3x_{167}\\
R_{120} &= -x_{143}x_{169} - x_{169} + x_{170}\\
R_{121} &= 2x_{168} + x_{168}^2 + x_{169}^2 - x_{170}^2\\
R_{122} &= -1 - x_{102}x_{169} + x_{112} + x_{159}x_{171} - x_{168} + x_{170}\\
R_{123} &= -x_{143}x_{173} - x_{173} + x_{174}\\
R_{124} &= 2x_{172} + x_{172}^2 + x_{173}^2 - x_{174}^2\\
R_{125} &= -1 - x_{102}x_{173} + x_{115} + x_{159}x_{175} - x_{172} + x_{174}\\
R_{126} &= -x_{143}x_{177} - x_{177} + x_{178}\\
R_{127} &= 2x_{176} + x_{176}^2 + x_{177}^2 - x_{178}^2\\
R_{128} &= -1 - x_{102}x_{177} + x_{118} + x_{159}x_{179} - x_{176} + x_{178}\\
R_{129} &= -x_{143}x_{181} - x_{181} + x_{182}\\
R_{130} &= 2x_{180} + x_{180}^2 + x_{181}^2 - x_{182}^2\\
R_{131} &= -1 - x_{102}x_{181} + x_{121} + x_{159}x_{183} - x_{180} + x_{182}\\
R_{132} &= -x_{143}x_{185} - x_{185} + x_{186}\\
R_{133} &= 2x_{184} + x_{184}^2 + x_{185}^2 - x_{186}^2\\
R_{134} &= -1 - x_{102}x_{185} + 2x_{102}x_{187} + x_{122} - 2x_{143}x_{187} + x_{159}x_{187}\\
&\quad - x_{184} + x_{185} + x_{186} - 3x_{187}
\end{aligned}
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\begin{aligned}
R_{135} &= -x_{143}x_{189} - x_{189} + x_{190} \\
R_{136} &= 2x_{188} + x_{188}^2 + x_{189}^2 - x_{190}^2 \\
R_{137} &= -1 - x_{102}x_{189} + x_{125} + x_{159}x_{191} - x_{188} + x_{190} \\
R_{138} &= -x_{143}x_{193} - x_{193} + x_{194} \\
R_{139} &= 2x_{192} + x_{192}^2 + x_{193}^2 - x_{194}^2 \\
R_{140} &= -1 - x_{102}x_{193} + x_{128} + x_{159}x_{195} - x_{192} + x_{194} \\
R_{141} &= 1 - x_{143}x_{197} - x_{197} + x_{198} \\
R_{142} &= -1 + 2x_{196} + x_{196}^2 + x_{197}^2 - 2x_{198} - x_{198}^2 \\
R_{143} &= -x_{102}x_{197} + 2x_{102}x_{199} + x_{129} - 2x_{143}x_{199} + x_{159}x_{199} - x_{196} \\
&\quad + x_{197} + x_{198} - 3x_{199} \\
R_{144} &= -x_{143}x_{201} - x_{201} + x_{202} \\
R_{145} &= 2x_{200} + x_{200}^2 + x_{201}^2 - x_{202}^2 \\
R_{146} &= -1 - x_{102}x_{201} + x_{133} + x_{159}x_{203} - x_{200} + x_{202} \\
R_{147} &= -x_{143}x_{205} - x_{205} + x_{206} \\
R_{148} &= 2x_{204} + x_{204}^2 + x_{205}^2 - x_{206}^2 \\
R_{149} &= -1 - x_{102}x_{205} + x_{136} + x_{159}x_{207} - x_{204} + x_{206} \\
R_{150} &= -x_{143}x_{209} - x_{209} + x_{210} \\
R_{151} &= 2x_{208} + x_{208}^2 + x_{209}^2 - x_{210}^2 \\
R_{152} &= -1 - x_{102}x_{209} + x_{139} + x_{159}x_{211} - x_{208} + x_{210} \\
R_{153} &= -6 - 2x_0 - 3x_0^2 - 2x_0x_4 - x_1^2 - x_2^2 \\
&\quad - 2x_4 - 2x_4^2 - x_5^2 - x_6^2 + 2x_{10} - x_{10}^2 \\
&\quad - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 \\
&\quad - x_{17}^2 - x_{18}^2 - x_{19}^2 - x_{100}^2 + 2x_{102} - 2x_{102}^2 \\
&\quad - x_{103}^2 - x_{104}^2 - x_{108}^2 - x_{109}^2 - x_{112}^2 - x_{115}^2 \\
&\quad - x_{118}^2 - x_{121}^2 - x_{122}^2 - x_{125}^2 - x_{128}^2 - x_{129}^2 \\
&\quad - x_{133}^2 - x_{136}^2 - x_{139}^2 + x_{142} - x_{212}^2 \\
R_{154} &= -x_5x_{10} - x_5x_{11} - x_5x_{12} - x_5x_{13} - x_5x_{14} - x_5x_{15} \\
&\quad - x_5x_{16} - x_5x_{17} - x_5x_{18} + x_{212} \\
R_{155} &= -16 + x_0 - x_{213}^2 \\
R_{156} &= -1 + x_2 + x_6 - x_6^2 \\
R_{157} &= -x_{143}x_{215} - x_{215} + x_{224} \\
R_{158} &= -1 + x_{214}^2 + x_{215}^2 - x_{224}^2 \\
R_{159} &= -x_{217} - x_{217}x_{218} + x_{225} \\
R_{160} &= -1 + x_{216}^2 + x_{217}^2 - x_{225}^2 \\
R_{161} &= -x_{209}^2 + x_{226} \\
R_{162} &= x_{215} - 2x_{220}x_{226} - x_{226} \\
R_{163} &= -4x_{215} - 8x_{215}x_{219} + x_{218} \\
R_{164} &= -x_{143} - x_{214}x_{221} + x_{218} \\
R_{165} &= -1 - x_{208} - x_{214}x_{222} + x_{216} \\
R_{166} &= -x_{100} - 4x_{209}x_{223} + x_{217} \\
R_{167} &= 2x_0^2 + 2x_0x_4 - 2x_0x_{148} - 2x_0x_{157} + x_1^2 - 2x_1x_{197} \\
&\quad + x_2^2 - 2x_2x_{185} + x_4^2 - 2x_4x_{157} + x_{10}^2 - 2x_{10}x_{163} \\
&\quad + x_{11}^2 - 2x_{11}x_{169} + x_{12}^2 - 2x_{12}x_{173} + x_{13}^2 - 2x_{13}x_{177} \\
&\quad + x_{14}^2 - 2x_{14}x_{181} + x_{15}^2 - 2x_{15}x_{189} + x_{16}^2 - 2x_{16}x_{193} \\
&\quad + x_{17}^2 - 2x_{17}x_{201} + x_{18}^2 - 2x_{18}x_{205} + x_{100}^2 - 2x_{100}x_{209} \\
&\quad - x_{143}x_{227} + x_{147}^2 + x_{148}^2 + x_{156}^2 + x_{157}^2 + x_{162}^2 \\
&\quad + x_{163}^2 + x_{168}^2 + x_{169}^2 + x_{172}^2 + x_{173}^2 + x_{176}^2 \\
&\quad + x_{177}^2 + x_{180}^2 + x_{181}^2 + x_{184}^2 + x_{185}^2 + x_{188}^2 \\
&\quad + x_{189}^2 + x_{192}^2 + x_{193}^2 + x_{196}^2 + x_{197}^2 + x_{200}^2 \\
&\quad + x_{201}^2 + x_{204}^2 + x_{205}^2 + x_{208}^2 + x_{209}^2 \\
R_{168} &= -x_{104}x_{228} + x_{129}
\end{aligned}
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\begin{aligned}
R_{169} &= -x_{109}x_{229} + x_{129} \\
R_{170} &= -x_{122}x_{230} + x_{129} \\
R_{171} &= -1 - x_3x_5 + x_5 + x_6 - x_{231}^2 - x_{232}^2 \\
&\quad - x_{233} - x_{233}^2 \\
R_{172} &= x_3x_5 - x_6 - x_{234}^2 - x_{235}^2 - x_{236} - x_{236}^2 \\
R_{173} &= -2 - x_{100}^2 - x_{148}^2 - x_{157}^2 + 2x_{163} - x_{163}^2 \\
&\quad - x_{169}^2 - x_{173}^2 - x_{177}^2 - x_{181}^2 - x_{185}^2 - x_{189}^2 \\
&\quad - x_{193}^2 - x_{197}^2 - x_{201}^2 - x_{205}^2 + x_{209}^2 - x_{237}^2 \\
&\quad - x_{238}^2 - x_{239} - x_{239}^2 \\
R_{174} &= -2x_{106}^2 - 2x_{111}^2 - 2x_{114}^2 - 2x_{117}^2 - 2x_{120}^2 - 2x_{124}^2 \\
&\quad - 2x_{127}^2 + x_{129}^2 - 2x_{131}^2 - 2x_{135}^2 - 2x_{138}^2 - 2x_{141}^2 \\
&\quad - x_{240}^2 - x_{241}^2 - x_{242} - x_{242}^2 \\
R_{175} &= -2 + x_{102} - 4x_{243} - 4x_{243}^2
\end{aligned}
$$